

Liquidity Fluctuations in Over-the-Counter Markets

VINCENT MAURIN

ABSTRACT

This paper proposes a theory of excess price fluctuations in over-the-counter secondary markets. When heterogeneous assets trade under asymmetric information, a quality effect emerges: high liquidity lowers the quality of the pool of sellers and decreases future liquidity. Cyclical equilibria can arise even without fundamental shocks. In a cycle, investors speculate by bidding up the price of low-quality assets, anticipating a high resale price at the peak. When this resale effect is strong, cycles disappear and multiple steady states coexist with different levels of liquidity. The model rationalizes empirical patterns for corporate bonds and housing in particular.

ASSET PRICES OFTEN APPEAR DISCONNECTED from fundamentals. This weak relationship shows as excess price fluctuations, documented notably for corporate bonds and housing by Bao and Pan (2013) and Glaeser et al. (2014). Price deviations from fundamentals are sometimes extreme during speculative episodes when prices rapidly increase before crashing (see Cutler, Poterba, and Summers 1991). In some asset classes, such as housing, the repeated nature of these episodes is puzzling. Why can price and trading volume fluctuate so much in the absence of changes to fundamentals?

This paper proposes a theory of endogenous liquidity fluctuations that can rationalize these phenomena. Motivated by the markets for corporate bonds

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Correspondence: Vincent Maurin, Stockholm School of Economics; address: Swedish House of Finance, Drottninggatan 98, 111 60 Stockholm, Sweden; e-mail: vincent.maurin@hhs.se

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and housing, I consider a secondary market in which heterogeneous assets trade under asymmetric information.¹ Asset owners face idiosyncratic liquidity shocks that generate gains from (re)trade. Although the supply of each asset quality is fixed, I show that the average quality of assets for sale and thus current liquidity are negatively related to past liquidity. This endogenous quality effect can generate cyclical fluctuations in equilibrium. In a cycle, price dynamics are speculative due to a resale effect. If future liquidity is high, forward-looking buyers who expect to easily resell a low-quality asset bid up its price today. This speculative behavior leads to endogenous asset price booms and crashes. When cycles do not exist, these resale considerations instead generate multiple equilibria.

To describe the quality effect and the resale effect, I turn to the model. The framework is stylized but is best applied to opaque over-the-counter (OTC) markets with heterogeneous assets, such as housing or corporate bonds. In the model, investors trade assets available in fixed supply. Each asset is characterized by its quality, called high or low, and each investor holds zero or one unit. Each period, with some probability, an asset owner experiences an idiosyncratic liquidity shock that captures a portfolio rebalancing need. In the model, a liquidity-hit investor has a lower private value from holding the asset. An investor with a low valuation can gain from selling his asset to a buyer with a high valuation. Buyers and sellers meet bilaterally in a decentralized market as in Duffie, Garleanu, and Pedersen (2005). The key friction is that only the current owner knows the quality of his asset and his own private valuation. This learning-by-holding assumption is meant to capture opacity of OTC markets in a tractable way.

In a bilateral OTC meeting, the buyer (she) makes the seller (he) a take-it-or-leave-it offer for the asset. The buyer faces adverse selection because she is uninformed about the asset's quality and the private valuation of the seller. She offers either a high price, or a low price, or she randomizes between both. The high price targets investors who own a high-quality asset and wish to sell because they have a low valuation. Under asymmetric information, owners of low-quality assets can also trade at this price and earn information rents. To avoid paying these rents, the buyer can instead offer the low price, targeted at low-valuation sellers of low-quality assets, but she would forgo the gains from trade with sellers of high-quality assets.

Market liquidity is defined as the (mixing) probability that a buyer makes a high offer. If buyers make a low offer, some gains from trade for high-quality assets are not realized compared to the symmetric information case. Market liquidity increases with the average quality of assets for sale, and decreases with the information rents conceded to low-quality sellers when making a high offer. Both variables are endogenous in a dynamic model: The composition of the pool of sellers responds to past liquidity, and, with forward-looking investors, information rents depend on future liquidity.

¹ See Bessembinder, Maxwell, and Venkataraman (2006) for evidence of opacity in the corporate bond market.

The first dynamic force at play in this opaque OTC market is a quality effect. The total supply of each asset quality is fixed but the share held by low-valuation investors is endogenous. This observation is crucial because high-quality assets may only be supplied by low-valuation investors as high-valuation owners cannot gain from trade. Suppose then most owners of high-quality assets have a low valuation. Buyers, who face a favorable seller pool, offer a high price. Next period, high-quality assets are held by these high-valuation buyers who do not sell unless they experience a liquidity shock. Faced with a less favorable pool, new buyers offer a low price: the market becomes illiquid. In an illiquid market, the potential supply of high-quality assets gradually increases as owners experience liquidity shocks but do not sell because buyers offer a low price. Eventually, the potential supply can be high enough for buyers to offer a high price again. The existence of the quality effect hinges only on the assumption that liquidity shocks are persistent.

The forward-looking behavior of investors who can buy and resell assets determines price dynamics in this secondary market. There exists a resale effect in that future market liquidity increases the value of low-quality assets. This value combines the holding value of the asset and the option value to resell it in the future. High future prices raise the resale value of a low-quality asset and thus its current value to a potential buyer, reducing information rents today. This resale effect implies that buyers are more likely to make a high-price offer if future buyers also make a high-price offer. The effect is stronger if investors are more forward-looking as the option to (re)sell is a larger component of the total value.

The interaction between the quality effect and the resale effect determines the liquidity dynamics of opaque secondary markets. With few plausible ingredients, the model can deliver a rich set of stationary equilibria that I fully characterize. A simple stability criterion is used to refine the equilibrium set in the case of multiplicity. The key parameter of interest is the (fixed) fraction of high-quality assets, which proxies for market quality.

The equilibrium is straightforward for extreme values of market quality. If it is very high, buyers always find it optimal to make a high offer despite the quality effect: the equilibrium is a liquid steady state. Similarly, if market quality is poor, the equilibrium is an illiquid steady state in which buyers always make a low offer. More interesting dynamics emerge when market quality is intermediate. The type and properties of the equilibrium then depend on the strength of the quality effect relative to the resale effect.

When the discount factor of investors and the probability of a liquidity shock are low, the quality effect is strong. For intermediate values of market quality, the only stable equilibrium is a cycle with period $T \geq 2$.² A T -period cycle consists of $T - 1$ trough periods when only low-quality assets trade and one peak period when buyers make a high offer. Trading at the peak clears the market

² This cycle coexists with a steady state in which buyers mix between a high offer and a low offer each period with constant probability. But this steady state is unstable.

for high-quality assets, leading to the trough. In the trough, the pool quality improves until the next peak, due to the quality effect. While only low-quality assets are traded in the trough, the price increases to reflect the option value to later resell at the peak. Hence, due to the resale effect, price dynamics in opaque OTC markets can be speculative.

The length of the cycle is decreasing with market quality. If market quality is low, it takes longer for the pool to improve enough to justify a high offer by buyers at the peak. The analysis also shows that longer cycles are harder to sustain because the resale effect acts as a countervailing force. In a cycle, buyers must find it optimal to make a high-price offer at the peak. While the pool of sellers is most favorable, information rents are also maximal: The next peak is T periods away, as opposed to one period away before the peak. For a longer cycle, the seller pool improvement is weaker while the negative effect due to the resale effect is stronger. As a result, the more investors are forward-looking, the shorter the equilibrium cycle is.

When investors' discount factor is high, the resale effect is very strong and no cycles, even short ones, exist in equilibrium. Instead, because market liquidity is driven mostly by investors' expectations about future market liquidity, two stable equilibria coexist: a high-price steady state and a low-price steady state. Intuitively, if future buyers make a high offer, current buyers also make a high offer because information rents are low. Conversely, if future buyers make a low offer, current buyers are more inclined to make a low offer. Hence, liquidity is self-fulfilling in opaque secondary markets when resale considerations dominate (see Moore (2013)).³

To assess the cost of liquidity fluctuations in OTC markets, I compare the two-period cycle with the coexisting steady state in which buyers randomize between a high and a low offer. This steady-state equilibrium does not survive the stability analysis, but it provides a clean benchmark for the cyclical equilibrium. Under risk-neutrality, social welfare is maximized when the average misallocation of high-quality assets is minimized. In the two-period cycle, liquidity is high precisely when the effective supply of high-quality assets is high, thereby reducing average misallocation. This result implies that the cycle can dominate the steady state. However, another force plays against cycles when market quality is high. Market liquidity increases with market quality in the steady state, and misallocation decreases. In a cycle, the average probability of a high offer stays constant at $\frac{1}{2}$ and misallocation does not vanish. The increase in market quality translates instead into higher buyer profits at the peak, with no effect on liquidity or welfare.

The model offers stark predictions for price dynamics when the equilibrium is cyclical. Asset prices fluctuate in excess of fundamentals and can be speculative. These predictions are related in the next paragraphs to empirical patterns

³ In this parameter region, Asriyan, Fuchs, and Green (2019) show that sunspot equilibria can exist in which the economy fluctuates between the high-liquidity steady state and the low-liquidity steady state. Hence, excess price fluctuations can also be an equilibrium outcome in this case, but fluctuations are driven by sunspots rather than the quality effect.

in two markets motivating this analysis: corporate bonds and housing. Given that the theoretical analysis highlights the importance of the quality effect for cycles, the results are best applied to a pool of similar assets rather than a single asset in isolation. In the corporate bond market, one can interpret the assets as a pool of securities with the same ratings but hard-to-observe variation in credit quality.

The first prediction of the model is that asset prices fluctuate in excess of fundamentals. In the model, equilibrium fluctuations can arise although asset fundamentals are constant. As such, the results provide a liquidity-based rationale for excess price fluctuations in corporate bonds, which is documented by Bao and Pan (2013). That liquidity fluctuations contribute to (excess) price movements in this market is supported empirically by Bai, Bali, and Wen (2019). They show that a corporate bond liquidity factor exhibit strong fluctuations that are not explained by traditional equity factors. In addition, as rising prices ultimately fall in a cycle, the model predicts reversals in corporate bond prices. This result is supported by Bao, Pan, and Wang (2011) and corroborates their interpretation of price reversals as an illiquidity proxy.

The model also provides a tractable theory of speculative behavior with repeated booms and busts. In the trough of a cycle, buyers speculate by acquiring low-quality assets at low prices to later resell them at a high price at the peak. Prices build up and then revert once high-quality assets also trade. Cyclical equilibrium prices thus exhibit all of the features that Cutler, Poterba, and Summers (1991) associate with speculative dynamics identified in many asset classes. The model further predicts that prices accelerate in the boom phase, in line with the characterization of bubbles by Greenwood, Shleifer, and You (2019). Finally, the expected holding time of an asset decreases as the bubble grows, a pattern observed by Zwick, Nathanson, and DeFusco (2020) during the 2000 to 2005 housing boom.

Relation to the Literature. This paper contributes to a large literature on dynamic adverse selection. To the best of my knowledge, it is the first to derive a tractable model of liquidity cycles based on the quality effect. Several studies extend the framework of Akerlof (1970) by letting agents trade over several periods but without retrade (see Janssen and Roy (2002), Camargo and Lester (2014), Fuchs and Skrzypacz (2015), Moreno and Wooders (2016), and Kim (2017)). In these models, time on the market acts as a screening device: lemons trade first, market quality improves, and good assets eventually trade. Zryumov (2015) and Hu (2018) embed these dynamics in a model of entrepreneurial finance with aggregate productivity shocks and entry. In good times, all entrepreneurs enter and immediately receive financing at a pooling rate. In bad times, the screening mechanism described above is at work. My model differs from these papers in two important ways: the transition between regimes is purely endogenous, and the resale effect drives price dynamics.⁴

⁴ In these papers, as in Janssen and Karamychev (2002), entry partially replicates liquidity shocks in my model but the composition of “entrants” is exogenous.

Relatedly, Eisfeldt (2004), Kurlat (2013), and Bigio (2015) present macro models in which liquidity varies because of aggregate shocks, that is, fluctuations are still fundamental.

My work is closest to Chiu and Koepl (2016) and Asriyan, Fuchs, and Green (2019). As in my model, investors learn by holding, assets are retraded, and trading history is unobservable. Both of these papers also find that steady states with different degrees of liquidity coexist with a strong resale effect.⁵ Asriyan, Fuchs, and Green (2019) use this multiplicity to build stochastic equilibria driven by sunspots, which I do not consider. Instead, I characterize predictable cycles generated by the quality effect. This effect is not active in Asriyan, Fuchs, and Green (2019), who assume that liquidity shocks are i.i.d. Chiu and Koepl (2016) also consider persistent shocks, but they do not characterize cycles. In my model, if a cycle exists, which requires parametric restrictions, it is selected by a stability criterion over the steady state.⁶

Relatedly, Inderst (2005) studies a matching market with adverse selection and entry applied to the labor market. As in this paper, the composition of the pool of sellers/workers is endogenous, but only steady states are considered. Guerrieri and Shimer (2014) and Chang (2018) study an environment similar to mine, but use competitive search instead of random search. Because their equilibrium is typically separating, the composition of the pool of sellers is irrelevant in their model and the resale effect is muted. This paper also relates to the new monetarist literature in which assets are used as a means of exchange (see Lagos, Rocheteau, and Wright (2017) for a survey). Building on Velde, Weber, and Wright (1999), studies like Rocheteau (2011) and Bajaj (2018) consider a setup with asymmetric information about asset quality. In these models, an asset owner is a borrower by construction. In the language of my model, all asset owners are hit by a liquidity shock, a feature that shuts down any quality effect. Interestingly, cycles can still arise in these models but for very different reasons. Rocheteau and Wright (2013) show that a high future asset price relaxes today's credit constraints and thus decreases today's asset liquidity premium, generating endogenous fluctuations in this premium. Several studies analyze government interventions in markets with adverse selection. Philippon and Skreta (2012) or Tirole (2012) consider a static setup, while Fuchs and Skrzypacz (2015) consider a dynamic setup without resale.

The rest of the paper is organized as follows. Section I describes the model. Section II analyzes the case of myopic investors with a zero discount factor.

⁵ Hellwig and Zhang (2013) and Zhou (2019) show that endogenous information acquisition by buyers is yet another force generating equilibrium (steady state) multiplicity. Rather than the intertemporal complementarity at play here, Malherbe (2014) highlights an intratemporal complementarity in investors' portfolio choice between cash and an opaque asset that also results in multiplicity.

⁶ This model differs from theirs in other ways: mine is set in discrete time, has no matching frictions, and allows the low-quality asset to have value. The analysis of cycles is simpler in discrete time, a point made by Albrecht, Postel-Vinay, and Vroman (2013) in related work. Finally, gains from trading the low-quality asset are important to obtain meaningful price dynamics, as cycles would otherwise feature no trade in a trough.

This limit case delivers the key intuition for the existence of cycles based on the quality effect. Section III introduces the resale effect with forward-looking investors and Section IV describes the equilibrium in this general case. Section V contains empirical predictions for the corporate bond market and the housing market. Section VI studies the welfare properties of cycles. Section VII concludes. Proofs are in Appendix A.

I. The Model

Time is discrete, indexed by t , and runs forever. The economy is populated by risk-neutral and infinitely lived investors with discount factor $\delta < 1$. There is a fixed supply of heterogeneous assets normalized to one unit with two qualities, (H)igh or (L)ow, with respective shares q and $1 - q$. Throughout the paper, I refer to q as market quality. Each asset is indivisible, and investors may hold either zero or one unit.⁷ Asset holders have either a (h)igh valuation for assets or a (l)ow valuation. Investors who do not own an asset have a high valuation. The utility flow from holding an asset depends on the asset quality and on the private valuation of the owner. These utility flows are denoted by u_a^i , where $i \in \{l, h\}$ stands for the private valuation and $a \in \{L, H\}$ for the asset quality. Utility flows are ordered as follows.

ASSUMPTION 1 (Gains from Trade):

$$u_L^l \leq u_L^h < u_H^l < u_H^h. \quad (1)$$

For both qualities, there are gains from trade between low-valuation and high-valuation investors. The inequality $u_L^h < u_H^l$ further implies that an H asset is more valuable than an L asset, regardless of the owner's private valuation. This last condition is not crucial for the results, as will become clear. I call a type ia investor an asset owner with valuation $i \in \{l, h\}$ and asset quality $a \in \{L, H\}$. Each period, an asset owner with valuation h is hit by a liquidity shock with probability $\gamma \in (0, 1)$. This shock is independently distributed across asset owners and is persistent: the investor has a low valuation l until he sells the asset.⁸ Nonasset owners that have a high valuation are endowed with e units of a numeraire good. The endowment allows a buyer to purchase any asset from any owner type.

ASSUMPTION 2 (Large Endowment):

$$e \geq \frac{u_H^h}{1 - \delta}. \quad (2)$$

⁷ The indivisibility comes without loss of generality with linear preferences because lotteries can be used. See also footnote 12.

⁸ In a previous version of this paper, the private valuation of an investor (high or low) switched with a given probability independent of his asset-holding status. All of the qualitative results hold under this alternative specification. See the related discussion in Chiu and Koepl (2016).

The key trading friction is that both the quality of the asset held by an investor and his valuation are the investor's private information.⁹ Trading takes place in a decentralized market with bilateral random matching, the OTC market. Each period, each asset owner is matched with a buyer drawn from a pool of nonasset owners of size larger than one. If trade occurs, the seller leaves the asset market forever, while if trade fails, the buyer leaves the market forever.¹⁰ In [Internet Appendix IA.II](#), I analyze the case in which asset owners face matching frictions.

In a match, the uninformed buyer (she) makes a take-it-or-leave-it offer for the asset of the informed seller (he).¹¹ In this screening game, the buyer assigns belief $\mu_{a,t}^i$ that the asset owner has type ia . Under random matching, the rational belief $\mu_{a,t}^i$ is the share of type ia among asset owners, and I denote by $\mu_t = \{\mu_{a,t}^i\}_{a=L,H}^{i=L,H}$ the distribution across types in period t . The strategy of a buyer who matches with a seller in period t is a distribution F_t over prices given beliefs μ_t .¹² A seller's strategy is an acceptance decision given the price offer. The solution concept for the bilateral trading game is the perfect Bayesian equilibrium.

Discussion of the Assumptions. As will become clear, the results hinge on three key ingredients of the model: an asset quality is only known to the current owner, liquidity shocks that generate trade are persistent, and trading history is unobservable.

Learning by Holding. This simplifying assumption is also made by Chiu and Koepl (2016) and Asriyan, Fuchs, and Green (2019) to ensure tractability. For real assets such as housing, Martel, Mirkin, and Waters (forthcoming) document price patterns suggestive of learning by holding. Plantin (2009) observes that some financial assets entitle their owner to superior information. According to Ivashina and Sun (2011), syndicated lenders routinely receive private information from corporate borrowers during loan amendments. In a market

⁹ Although qualitative differences emerge, whether liquidity shocks are privately or publicly observed is not crucial for the result. The key difference is that h -valuation owners are unable to sell L assets if shocks are publicly observable, and thus, adverse selection is reduced. [Internet Appendix IA.I](#) shows that liquidity cycles still arise, but for lower values of market quality. The [Internet Appendix](#) is available in the online version of the article on *The Journal of Finance* website.

¹⁰ This assumption avoids keeping track of the value function of nonasset owners. The analysis is similar if investors stay in the market, as in a previous version of this work.

¹¹ If instead the informed seller makes the offer, the trading game becomes a signaling game. [Internet Appendix IA.II](#) shows that the perfect Bayesian equilibrium of the signaling game is similar under the undefeated equilibrium concept of Mailath, Okuno-Fujiwara, and Postlewaite (1993).

¹² The equilibrium of this trading game is identical when buyers can offer richer contracts specifying a price and a probability to trade. In that case, a strategy is a two-dimensional distribution over prices and trading probabilities. Samuelson (1984) shows that with linear preferences, buyers always make a single offer with probability of trade equal to 1. This can be implemented with price offers.

with heterogeneous assets such as corporate bonds, a buyer may need to evaluate many assets prior to trading. It is therefore plausible that the seller that owns the asset has superior information.

Persistent Liquidity Shocks. Liquidity shocks capture household relocation needs. For financial assets, liquidity shocks can be interpreted as hedging needs due to changes in the valuation of other portfolio assets or new private investment opportunities. With these interpretations in mind, it is reasonable to assume that the liquidity shock persists until the asset is sold. An investor that faces tight funding constraints or concentration limits may be unable to undertake new investments before old positions are closed.

Opacity of Trade History. With observable trade history, a buyer could infer the quality of an asset from its trade history if assets of different quality exhibited different trading patterns. When trade history is opaque, this screening tool is not available. This assumption makes it easier for low-quality asset buyers to resell their assets, and, as I discuss later, is necessary for cycles to emerge.¹³ Empirically, in OTC markets, assets transit through dealers before reaching end customers (see Hollifield, Neklyudov, and Spatt (2017) for asset-backed securities or Li and Schürhoff (2019) for municipal bonds). Hence, trading history is difficult to obtain. The ability of financial intermediaries to pool and tranche securities in “kitchen-sink” bonds or barbell-shaped loan pools also contributes to the opacity of assets’ trading history.

II. Myopic Liquidity Cycles

In this section, I analyze the special case in which $\delta = 0$, that is, investors are not forward-looking. This limit case delivers the key intuition for the existence of cycles based on the quality effect. Setting $\delta = 0$ simplifies the problem: equilibria have a Markov structure, and the equilibrium stability analysis is straightforward. I turn to the analysis of richer price dynamics in Section III. Below, I first analyze the trading game between the buyer and the seller. I then explain the dynamics of the model, which, in this limit case, are only backward-looking. In a bilateral match, the buyer makes a take-it-or-leave-it price offer to the seller. The optimal strategy for seller ia is to accept any price weakly above his holding value u_a^i . Under asymmetric information, the buyer anticipates that offer u_a^i is accepted not only by type ia but also by inframarginal types with a lower holding value. It is optimal for a buyer to offer either a low price u_L^l , which is accepted only by type lL , or a high price u_H^l , which is accepted by type lH as well as by types hL and lL under Assumption 1.¹⁴ The buyer’s

¹³ Hendel, Lizzeri, and Siniscalchi (2005) and Kim (2017) study the equilibrium effects of the observability of past trade history in a dynamic lemon market without resale.

¹⁴ Offer u_a^h is strictly dominated by offer $u_a^l < u_a^h$ for $a \in \{L, H\}$. Increasing the offer to u_a^h from u_a^l strictly decreases profit for types lower than ha , and new trades made with ha owners yield zero profit because the buyer also has high valuation h .

utility from a low-price offer u_L^l and a high-price offer u_H^l is, respectively,

$$v_t^B(u_L^l) := \mu_{L,t}^l(u_L^h - u_L^l), \quad (3a)$$

$$v_t^B(u_H^l) := \mu_{H,t}^l(u_H^h - u_H^l) - (1 - q)(u_H^l - u_L^h). \quad (3b)$$

With a high offer, the buyer captures the gains from trade with low-valuation sellers of H assets, but she pays a premium for L assets. With a low offer, the buyer forgoes the gains from trade with H -asset sellers, but she trades the L asset at the lower reservation price of lL investors. With two possible offers, the buyer's strategy in period t can be described by the probability $\pi_t \in [0, 1]$ that a high-price offer u_H^l is made. This probability π_t is a measure of market liquidity because $1 - \pi_t$ is the fraction of valuable H asset trades that do not take place under asymmetric information.

Market liquidity π_t is determined by the distribution μ_t of assets. Equations (3a) and (3b) show that the net benefit from a high offer $v_t^B(u_H^l) - v_t^B(u_L^l)$ increases with $\mu_{H,t}^l$, the mass of H assets held by l investors. This variable is the effective supply of H assets because hL investors, who do not gain from trade, do not sell. As shown below, $\mu_{H,t}^l$ need not converge to a steady-state value in the long run as self-sustained fluctuations can arise in equilibrium.

I now derive the law of motion for the effective supply of H assets. At the beginning of period t , there is mass $\mu_{H,t}^l$ of lH investors. At the end of period t , this mass is equal to $(1 - \pi_t)\mu_{H,t}^l$, as a fraction π_t of these investors sells to h -valuation buyers. All other assets are held by h -valuation investors. Moving to period $t + 1$, a fraction γ of the high-valuation owners receive a liquidity shock and switch to l valuation. The law of motion follows

$$\mu_{H,t+1}^l = \gamma \left[\mu_{H,t}^h + \pi_t \mu_{H,t}^l \right] + (1 - \pi_t) \mu_{H,t}^l = \gamma q + (1 - \gamma)(1 - \pi_t) \mu_{H,t}^l, \quad (4)$$

using the market-clearing condition $\mu_{H,t}^l + \mu_{H,t}^h = q$ to substitute for $\mu_{H,t}^h$. Equation (4) shows that there is a quality effect whereby high market liquidity π_t reduces the future effective supply of H assets $\mu_{H,t+1}^l$. In a liquid market ($\pi_t = 1$), all H assets are held or acquired by h -valuation investors. Next period, the supply is therefore equal to its lowest possible value, $\mu_{H,t+1}^l = \gamma q$. Conversely, if the market is illiquid in period t ($\pi_t = 0$), the effective supply increases, that is, $\mu_{H,t+1}^l > \mu_{H,t}^l$, by law of motion (4). This supply build-up arises because all lH investors who did not trade in period t and hH investors who experienced a liquidity shock contribute to the supply next period.

No quality effect exists for L assets because lL investors trade with probability one. The mass of L assets held by lL investors is thus constant, with

$$\mu_{L,t}^l = \gamma(1 - q). \quad (5)$$

The solution of the trading game depends on the effective supply of H assets $\mu_{H,t}^l$, which is the only endogenous variable of the economy.

DEFINITION 1: An equilibrium is given by a value $\mu_{H,0}^l$, law of motion (4), and a probability π_t that a buyer makes the high offer u_H^l , where

$$\pi_t = \begin{cases} 0 & \text{if } \mu_{H,t}^l < \hat{\mu} \\ \pi \in [0, 1] & \text{if } \mu_{H,t}^l = \hat{\mu}, \\ 1 & \text{if } \mu_{H,t}^l > \hat{\mu} \end{cases} \quad (6)$$

$$\hat{\mu} := (1 - q) \frac{(u_H^l - (1 - \gamma)u_L^h - \gamma u_L^l)}{u_H^h - u_H^l}. \quad (7)$$

Equation (6) shows that a buyer makes the high offer u_H^l if and only if the effective supply of H assets exceeds a threshold $\hat{\mu}$. At this value, the expected gain from trading the H asset, given by $\hat{\mu}(u_H^h - u_H^l)$, equals the average rent paid to L asset owners with a high offer, given by the numerator of $\hat{\mu}$.

It is now possible to characterize stationary equilibria of the model. Observe that Definition 1 also characterizes transition dynamics from any initial value $\mu_{H,0}^l$. This feature is helpful to study equilibrium stability, that is, whether the economy converges back to the equilibrium after a small perturbation to the state variable μ_H^l .¹⁵ Before stating the result, I introduce the following thresholds for market quality, which is the main variable of interest:

$$\forall T \geq 2, \quad \bar{q}_T := \frac{u_H^l - (1 - \gamma)u_L^h - \gamma u_L^l}{[1 - (1 - \gamma)^{T-1}](u_H^h - u_H^l) + u_H^h - (1 - \gamma)u_L^h - \gamma u_L^l}. \quad (8)$$

and $\bar{q}_1 = \bar{q}_2$, $\bar{q}_\infty := \lim_{T \rightarrow \infty} \bar{q}_T$.

PROPOSITION 1: When $\delta = 0$, the unique stable stationary equilibrium is¹⁶

- i) an illiquid steady state with a low-price offer u_L^l by buyers if $q \in [0, \bar{q}_\infty]$,
- ii) a T -period cycle with a peak period with a high offer u_H^l and $T - 1$ trough periods with a low offer u_L^l by buyers if $q \in [\bar{q}_{T+1}, \bar{q}_T]$ for $T \geq 2$, and
- iii) a liquid steady state with a high-price offer u_H^l by buyers if $q \in [\bar{q}_1, 1]$.

The result is a special case of Proposition 2. The nature of the equilibrium is intuitive for extreme market quality q . Suppose q is high and buyers make the high offer u_H^l . The effective supply of H assets is then equal to $\mu_H^l = \gamma q$ by equation (4). If q is higher than $\bar{q}_1$, the effective supply exceeds the threshold $\hat{\mu}$. It is again optimal for buyers to offer the high price, so the equilibrium is a liquid steady state in which buyers always make the high offer. A similar argument can be used to determine the threshold $\bar{q}_\infty$ below which the equilibrium is the illiquid steady state in which buyers make the low offer u_L^l every period. These two extreme cases are illustrated in Figure 1. In Panel 1(a) (Figure 1e), the threshold $\hat{\mu}$ lies to the left (right) of the state space for μ_H^l , and thus, a high offer is always (never) optimal.

¹⁵ Appendix B provides a formal definition of the stability criterion.

¹⁶ The formal definitions of a steady state and a cycle are postponed to Section IV to avoid redundancies.

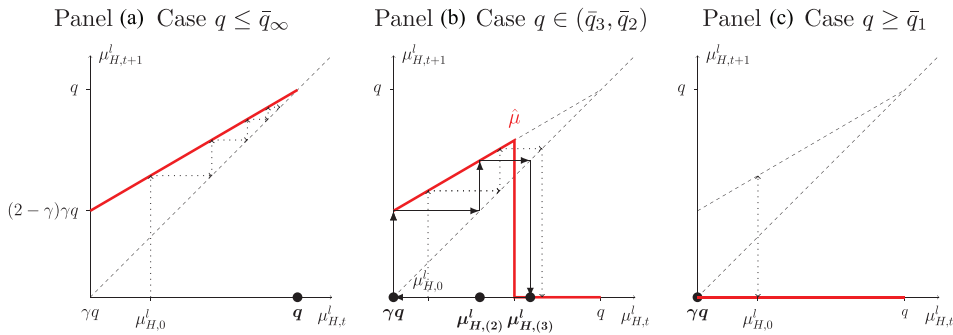


Figure 1. Equilibrium law of motion ($\delta = 0$). The very tick (red) line represents the equilibrium mapping for the state variable μ_H^l . The dots indicate the values of μ_H^l in the equilibrium state space. The thin solid black line with arrows in Panel B illustrates the equilibrium trajectory in the stable periodic cycle. The dotted line indicates convergence to the stationary equilibrium. (Color figure can be viewed at wileyonlinelibrary.com)

For intermediate market quality q , the equilibrium is a cycle. In this parameter region, the effective supply of H assets fluctuates across the liquidity threshold $\hat{\mu}$ because of the quality effect. When μ_H^l lies above $\hat{\mu}$, buyers make a high offer. By law of motion (4), the effective supply of H assets in the next period is equal to γq , which lies below the threshold $\hat{\mu}$ if $q \leq \bar{q}_1$. The market then becomes illiquid, that is, $\pi_{t+1} = 0$ as buyers now prefer to make the low offer u_L^l . The effective supply μ_H^l builds up as long as the market stays illiquid as more H asset owners are hit by a liquidity shock but do not sell. Eventually, if $q \geq \bar{q}_\infty$, μ_H^l exceeds the threshold $\hat{\mu}$ and the market is liquid again. Intuitively, it takes longer for the effective supply to cross the threshold $\hat{\mu}$ if market quality q is low. Hence, longer cycles exist for lower values of q , which is why $\bar{q}_T$ decreases with T . Cyclical dynamics are illustrated in the central panel of Figure 1 for a three-period cycle.

Finally, Proposition 1 shows that the unique stationary equilibrium is stable. This result is best understood graphically thanks to Figure 1. In particular, Panel 1(c) illustrates the case of a three-period cycle and shows that this equilibrium coexists with a steady-state equilibrium in which $\mu_{H,t}^l = \hat{\mu}$ for all t . In the (unstable) steady-state equilibrium, buyers randomize between the high offer and the low offer with a constant probability each period. To understand why the cycle is the stable equilibrium, consider a perturbation to a value $\mu_{H,0}^l$, to the left of $\hat{\mu}$, as in Panel 1(c). The effective supply of H assets builds up until buyers switch to a high-price offer. Convergence to the cycle then occurs because the effective supply is reset to its lowest value γq , which belongs to the state space of the cycle.

III. Liquidity Dynamics and Resale Effect

I now analyze the general case in investors' discount factor δ is strictly positive. The key difference relative to Section II is that investors' valuation for

the asset is now endogenous as it incorporates expectations about future market liquidity. The analysis below will show that equilibrium dynamics are also driven by a resale effect whereby current market liquidity increases with future market liquidity. As will become clear, the resale effect introduces a speculative component to asset prices, and it generates richer dynamics.

To account for the dynamic component in investors' valuation, let $r_{a,t}^i$ be type ia investor's value for the asset after the trading stage. By definition, $r_{a,t}^i$ is also the reservation price below which an ia investor would not sell his asset. Symmetrically, $r_{a,t}^h$ is the maximum a buyer is willing to pay for asset a under symmetric information. When $\delta = 0$, as in Section II, $r_{a,t}^i = u_a^i$, that is, the reservation price is equal to the one-period holding value. When $\delta > 0$, the reservation price also has a forward-looking component, which I define explicitly below. The asset owner's valuation before the trading stage can be derived from the reservation price as

$$v_{a,t}^i := \int_p \max \{r_{a,t}^i, p\} dF_t(p), \quad (9)$$

where F_t is the cumulative distribution function of price offers in period t . Compared to the reservation price $r_{a,t}^i$, the valuation $v_{a,t}^i$ accounts for the option to sell the asset. Anticipating the equilibrium of the trading game, an asset owner sells when the price offer p exceeds his reservation price $r_{a,t}^i$, which explains equation (9).

A. Trading Game

The first step of the analysis is the characterization of the outcome of the trading game between the buyer and the seller. Buyers have the bargaining power, so a type ia seller accepts any offer weakly above his reservation price $r_{a,t}^i$. The buyer thus anticipates that an offer p will be taken by all types with reservation price below p . Given the distribution of seller types μ_t , the buyer's expected profit is given by

$$v_{B,t} := \max_p \sum_{\substack{i \in \{l, h\} \\ a \in \{L, H\}}} \mu_{a,t}^i \mathbb{1}_{\{p \geq r_{a,t}^i\}} (r_{a,t}^h - p). \quad (10)$$

If she trades with type ia , the buyer's payoff is the difference between her valuation of the asset, $r_{a,t}^h$, and the price paid, p . To compute the buyer's expected profit from offer p , the probability weight for each type is equal to the share $\mu_{a,t}^i$ in the population multiplied by an indicator variable $\mathbb{1}_{\{p \geq r_{a,t}^i\}}$ that the offer is accepted by that type.

To solve for the trading game of Section II with $\delta = 0$, I use the ranking between holding values imposed by Assumption 1. The trading game with $\delta > 0$ is similar after replacing holding values u_a^i with endogenous reservation prices $r_{a,t}^i$. Lemma 1 shows that the ranking of types based on holding values in Assumption 1 is preserved.

LEMMA 1: Under the transversality condition $\lim_{t \rightarrow \infty} \delta^t r_{a,t}^i = 0$, for any t , the reservation prices of asset owners satisfy

$$r_{L,t}^l \leq r_{L,t}^h < r_{H,t}^l < r_{H,t}^h. \quad (11)$$

Lemma 1 states that a seller with a lower holding value also has a lower reservation price. Intuitively, the reservation price has two components: the present value of holding the asset forever and the option value of selling it in some future period. The option value component is the same across all investors because buyers make uninformed offers. However, the holding value component increases with the current utility flow, which explains why the static ranking of types according to utility flows is preserved in the dynamic model. The transversality condition needed for Lemma 1 is satisfied in equilibrium (see Definition 2).

The ranking of types in Lemma 1 implies that the adverse selection problem faced by buyers is the same as in Section II. Hence, a buyer makes either a high offer $r_{H,t}^l$, which targets lH investors and is also accepted by hL investors and lL investors, or a low offer $r_{L,t}^l$, which is only accepted by lL investors.

LEMMA 2 (Trading Game): A buyer makes either a low offer $r_{L,t}^l$ or a high offer $r_{H,t}^l$. For any t , the mixing probability π_t of a high offer is given by

$$\pi_t = \begin{cases} 0 & \text{if } \Gamma_t < 0 \\ \pi \in [0, 1] & \text{if } \Gamma_t = 0, \\ 1 & \text{if } \Gamma_t > 0 \end{cases} \quad (12)$$

$$\text{where } \Gamma_t := \underbrace{\mu_{H,t}^l}_{\text{Effective Supply of } H \text{ Assets}} (r_{H,t}^h - r_{H,t}^l) - (1 - q) \underbrace{\left[r_{H,t}^l - \frac{\mu_{L,t}^h}{1 - q} r_{L,t}^h - \frac{\mu_{L,t}^l}{1 - q} r_{L,t}^l \right]}_{\text{Average Information Rent}}. \quad (13)$$

As in Section II, buyers choose between a low price that targets lL investors and a high price that targets lH investors. When $\delta = 0$, the only endogenous component shaping the buyer trade-off is the effective supply $\mu_{H,t}^l$ of H assets. The key difference when investors are forward-looking is that reservation prices and thus trading prices, $r_{L,t}^l$ and $r_{H,t}^l$, are endogenous. From equation (13), the net benefit from a high offer Γ_t and market liquidity π_t increase with the gains from trading H assets, $r_{H,t}^h - r_{H,t}^l$, and decrease with the average information rent. As I show below, these new forces generate a dynamic resale effect.

B. Dynamics

Law of motion. By Lemma 2, an lL investor always trades, but an lH investor only trades with probability π_t when the buyer makes a high offer $r_{H,t}^l$. This

outcome of the trading game implies that the laws of motion for H assets and L assets are still given by equations (4) and (5), respectively. In particular, the quality effect operates exactly as in the case $\delta = 0$.

Reservation prices. To derive the reservation prices for H assets, note that an H -asset owner does not gain from trading because he never receives an offer strictly above his reservation price. Hence, he enjoys his autarkic utility, which is the discounted utility flow from holding the asset forever.

LEMMA 3: If $\lim_{t \rightarrow \infty} \delta^t r_{a,t}^i = 0$, an H asset owner enjoys utility

$$\forall t, \quad v_{H,t}^l = r_{H,t}^l = \frac{u_H^l}{1 - \delta} \quad (14a)$$

$$v_{H,t}^h = r_{H,t}^h = \frac{u_H^h + \delta \gamma r_{H,t}^l}{1 - \delta(1 - \gamma)}. \quad (14b)$$

Lemma 3 implies that the gains from trading an H asset, captured by the difference in reservation prices $r_{H,t}^h - r_{H,t}^l$, are constant. Hence, the average information rent is the only forward-looking term shaping buyers' trade-off characterized by equation (12).

The recursive equation for the reservation prices of L asset owners is

$$r_{L,t}^l = u_L^l + \delta [\pi_{t+1} r_H^l + (1 - \pi_{t+1}) r_{L,t+1}^l], \quad (15a)$$

$$r_{L,t}^h = u_L^h + \delta [\pi_{t+1} r_H^l + (1 - \pi_{t+1}) \{ (1 - \gamma) r_{L,t+1}^h + \gamma r_{L,t+1}^l \}]. \quad (15b)$$

An L asset owner strictly prefers to sell his asset when he receives a high-price offer r_H^l , which arises with probability π_{t+1} in period $t + 1$. In this case, he earns a trading profit $r_H^l - r_{L,t+1}^l$ equal to his information rent. Thus, unlike H -asset owners, L -asset owners can strictly benefit from the asset market if they receive a high offer with positive probability in the future.

Resale effect. It is now possible to explain how current liquidity increases with future liquidity via a resale effect. As explained before, the only forward-looking component determining market liquidity π_t is the average information rent to L asset owners, which can be decomposed using (15a) and (15b) as

$$r_{H,t}^l - (1 - \gamma) r_{L,t}^h - \gamma r_{L,t}^l = r_H^l - r_{L,t}^h - \gamma (r_{L,t}^h - r_{L,t}^l) \quad (16a)$$

$$= r_H^l - r_{L,t}^h - \gamma (u_L^h - u_L^l + \delta(1 - \gamma)(1 - \pi_{t+1})(r_{L,t+1}^h - r_{L,t+1}^l)). \quad (16b)$$

Future liquidity increases current liquidity via two channels, as captured by each term on the right-hand side of equation (16a). The first term is the difference between the high price r_H^l and a buyer's valuation for an L asset,

$r_{L,t}^h$. The higher π_{t+1} , the higher $r_{L,t}^h$ because the current buyer is more likely to resell the asset at a high price next period. This pure resale effect lowers information rents today. Intuitively, a buyer is less worried about adverse selection if he can resell a lemon at a high price.

The second term, $r_{L,t}^h - r_{L,t}^l$, which enters negatively, is the trading profit from a low offer. This profit decreases with future market liquidity π_{t+1} , all else equal, due to an intertemporal competition effect. If $\pi_{t+1} = 1$, both the lL asset owner and a potential h -valuation buyer can then sell at a high price in period $t + 1$. Because the buyer profit from a low offer is limited to the difference in current holding values, $u_L^h - u_L^l$, a high offer is relatively more attractive in period t . Hence, this intertemporal competition effect also implies that future liquidity begets current liquidity.

The pure resale effect and the intertemporal competition effect induce similar dynamics in liquidity. For simplicity, in what follows I use the term resale effect for both channels. Note that the dynamic competition effect is absent when $u_L^h = u_L^l$, which is sometimes analyzed as a limit case. This resale effect is strong if investors' discount factor δ is high.

C. Stationary Equilibrium Concept

The equilibrium analysis in the general case is centered on the long-run dynamics of the model. When $\delta > 0$, transition dynamics do not have a Markov structure with respect to the predetermined state variable μ_H^l . As discussed above, the solution to the buyer optimization problem also depends on reservation prices $r_{a,t}^i$, which are not predetermined and may depend on the entire future path of market liquidity $\{\pi_{t+k}\}_{k \geq 1}$. Hence, the focus is on (Markov) stationary equilibria in which fluctuations are driven only by changes in the predetermined state variable μ_H^l .

DEFINITION 2: A stationary equilibrium is given by a period T , a state space $\mathcal{M} = \{\mu_{H,t}^l\}_{t=1,\dots,T}$, a buyer mixing probability function $\pi(\cdot)$, and reservation price functions $\{r_a^i(\cdot)\}_{a=L,H}^{i=l,h}$ defined over $\mathcal{M}$ such that, for any $t \in [1, T]$, (with $T + 1 = 1$ by convention),

- i) $\pi_t = \pi(\mu_{H,t}^l)$ is given by (12) with $r_{a,t}^i = r_a^i(\mu_{H,t}^l)$,
- ii) $r_{L,t}^i = r_L^i(\mu_{H,t}^l)$ satisfies equations (15a) and (15b), and
- iii) $\mu_{H,t+1}^l$ is given by law of motion (4) with $\pi_t = \pi(\mu_{H,t}^l)$.

The cardinality T of the state space $\mathcal{M}$ is called the period of the equilibrium.

The first condition of Definition 2 formalizes the optimal choice of buyers between a high offer and a low offer. The second and third conditions respectively state that the sequence of reservation prices and the law of motion for the asset distribution must be consistent with the buyers' optimal choice to form part of an equilibrium. A stationary equilibrium is characterized by the state space for

the effective supply of H assets μ_H^l . Because there is no aggregate exogenous shock in this model, fluctuations over this state space are purely endogenous if they exist.

A stationary equilibrium is characterized by its period, that is, the smallest time interval T between two visits of a given value μ_H^l of the state space. I introduce the following taxonomy for stationary equilibria.

DEFINITION 3: A steady state is a stationary equilibrium with period $T = 1$. A cycle is a stationary equilibrium with period $T \geq 2$.

In a steady state, endogenous variables are constant over time. In a cycle, however, the economy fluctuates deterministically over a nondegenerate state space. As the analysis will show, there can be several stationary equilibria for some parameter values. To refine the equilibrium set, I use a stability criterion based on the standard concept of Lyapunov stability for discrete-time systems (see, for instance, Anishchenko, Vadivasova, and Strelkova (2014)). The underlying logic is intuitive: a stationary equilibrium is stable if, following any small initial perturbation of the state variable, there is an equilibrium path that converges back to this stationary equilibrium. In the interest of space, the formal definition of this criterion is given in Appendix B.

IV. Equilibria with Resale Effect

The analysis so far shows that two forces drive the equilibrium dynamics, namely, a quality effect and a resale effect. The quality effect implies that past liquidity crowds out current liquidity as it reduces the effective supply of H assets. Instead, the resale effect implies that future liquidity crowds in current liquidity due to the intertemporal complementarity in buyers' decisions. To streamline the presentation, I first consider the parameter space in which the resale effect is weaker, which is given by the condition

$$(1 - \delta - \gamma)(u_H^l - u_L^l) \geq \frac{(1 - \delta)^2(1 - \gamma)^2}{1 - \delta(1 - \gamma)}(u_L^h - u_L^l). \quad (C_1)$$

This condition has an intuitive interpretation when there are no gains from trading the L asset, that is, when $u_L^h = u_L^l$. Condition (C_1) then becomes $1 - \gamma \geq \delta$. In law of motion (4), the term $1 - \gamma$ is the marginal (negative) effect of liquidity on the future effective supply of H assets, that is, the strength of the quality effect. The discount factor δ captures the strength of the resale effect. When condition (C_1) holds, the quality effect dominates the resale effect.¹⁷

A. Weak Resale Effect

I analyze first the weak resale effect case, assuming (C_1) holds.

¹⁷ This simplified version of (C_1) is equivalent to the condition $\kappa > r$ of Chiu and Koepl (2016). They assume that the low-quality asset is a pure lemon, that is, $u_L^h = u_L^l = 0$.

PROPOSITION 2: *There exist market quality thresholds $\{q_T, \bar{q}_T\}_{T \geq 1}$ and discount factor thresholds $\{\bar{\delta}_T\}_{T \geq 2}$ such that under (C_1) , the following hold:*

1. *Steady State. There exists a unique steady state for each value of q such that market liquidity is given by $\pi^{ss} = 0$ for $q \leq \underline{q}_1$, $\pi^{ss}(q) \in (0, 1)$ for $q \in (\underline{q}_1, \bar{q}_1)$, and $\pi^{ss} = 1$ for $q \geq \bar{q}_1$.*
2. *Cycles. A period $T \geq 2$ cycle exists for $q \in [\underline{q}_1, \bar{q}_T]$ if and only if*

$$\delta \leq \bar{\delta}_T.$$

This cycle has $T - 1$ trough periods with buyers making the low offer, and one peak with buyers making the high offer with probability

$$\pi_0^{(T)} = \begin{cases} \pi_0^{(T)}(q) \in (0, 1) & \text{if } q \in (\underline{q}_1, \underline{q}_T) \\ 1 & \text{if } q \in [\underline{q}_T, \bar{q}_T]. \end{cases}$$

For $T \geq 2$, longer cycles exist for lower values of δ as $\bar{\delta}_T > \bar{\delta}_{T+1}$. Longer cycles without mixing exist for lower values of q as $\underline{q}_T > \bar{q}_{T+1}$.

The expressions for thresholds $\underline{q}_T$, $\bar{q}_T$, and $\bar{\delta}_T$ are given in the proof. Proposition 2 extends the results of Proposition 1 to the case in which condition (C_1) holds. When market quality is high (low), the equilibrium is a liquid (illiquid) steady state. When market quality is intermediate, cycles with one peak period and several trough periods emerge, but only if δ is low enough, that is, the resale effect is weak. Proposition 2 also shows that other equilibria exist when market quality is intermediate, notably a steady state in which buyers mix between offers with a constant probability. As shown below, only the cycle without mixing survives the stability criterion.¹⁸

As explained in Section II, cycles are generated by the quality effect. To avoid repetition, I focus here on the role of the resale effect. The key intuition for existence condition (C_T) is that the resale effect acts as a countervailing force on the quality effect. While the quality of the seller pool is best (worse) at the peak (trough) of the cycle, resale prospects follow an opposite pattern. Consider a buyer at the peak of a T -period cycle. The market will be illiquid in the next $T - 1$ periods, so information rents are highest at the peak. When information rents are high, a low offer is attractive, but the equilibrium play is a high offer. Conversely, in the period before the peak, information rents are lowest because the market is liquid in the next period. This makes it difficult to sustain a low offer in equilibrium before the peak because low information rents increase the attractiveness of a high offer.

Proposition 2 shows that a cycle exists only for a low enough discount factor δ . Furthermore, the longer the cycle, the lower the upper bound $\bar{\delta}_T$ on the

¹⁸ Proposition 2 characterizes all stationary equilibria for $T \leq 2$. For each $T \geq 3$, Proposition 2 characterizes the equilibrium in which buyers do not mix between offers. It also provides an existence condition for a cyclical equilibrium where buyers mix at the peak. This later equilibrium and other potential cycles of period $T \geq 3$ with mixing are shown to be unstable in Corollary 2.

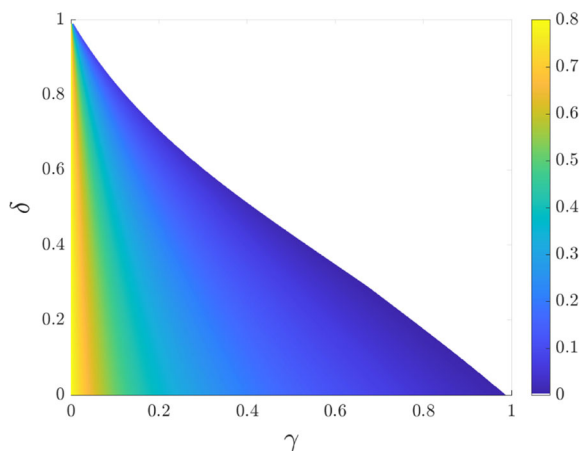


Figure 2. Existence region: two-period cycle. Valuation parameters $u_L^l = 1$, $u_L^h = 1.5$, $u_H^l = 2$, $u_H^h = 4$. The color bar maps each pair (γ, δ) to the interval size $[q_1, \bar{q}_1]$ over which a cycle exists. No cycle exists in the white region. (Color figure can be viewed at wileyonlinelibrary.com)

discount factor, that is, the more forward-looking investors are, the shorter the cycle is. Two effects explain why $\bar{\delta}_T$ is decreasing in T . First, the marginal improvement in pool quality during the trough decreases. In a T -period cycle without mixing by buyers, the effective supply of H assets is

$$\mu_{H,t}^{l,(T)} = [1 - (1 - \gamma)^t]q, \quad (17)$$

where $t \leq T - 1$ is a period in the trough. If the trough is long, most H -asset holders switch to a low valuation, so the marginal improvement in pool quality decreases. Second, at the peak, information rents are larger for longer cycles as a buyer must wait for the next peak to resell an asset at a high price.

Additional results about existence condition (C_T) can be obtained in the limit case with no gain from trading the low-quality asset.¹⁹

COROLLARY 1: *If $u_L^h = u_L^l$, existence condition (C_T) for cycles is tightened when γ increases, that is, $\bar{\delta}_T$ is decreasing in γ .*

Corollary 1 is intuitive. The parameter $1 - \gamma$ measures the strength of the quality effect. A strong quality effect can balance a strong resale effect, which is why cycles may exist for high values of the discount factor when γ is low. Figure 2 illustrates existence condition (C_T) for a two-period cycle. The upper limit of the colored region corresponds to the mapping $\gamma \mapsto \bar{\delta}_2$. Lighter shades of color correspond to larger intervals $[q_1, \bar{q}_1]$ of market quality where the two-period cycle exists. Figure 2 suggests that the cycle existence region expands as the probability of a liquidity shock γ decreases toward zero. In a two-period

¹⁹ This condition is not necessary for the result. An example is given in Figure 2, which obtains under the assumption $u_L^h > u_L^l$.

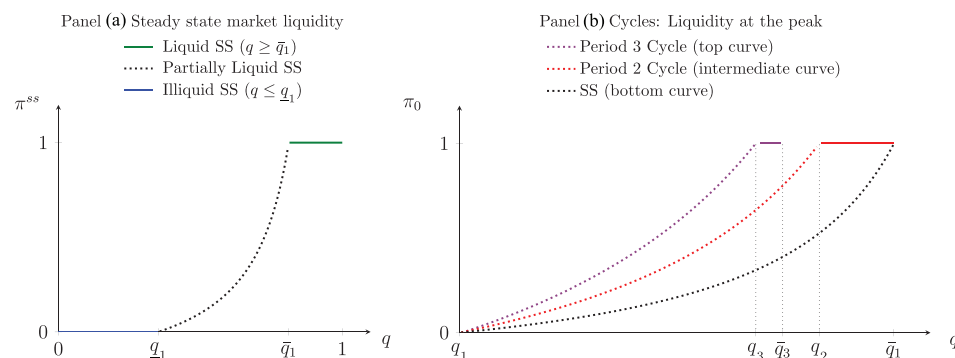


Figure 3. Equilibria when (C_T) holds for $T \leq 3$. Parameter values: discount factor $\delta = 0.2$, probability of a shock $\gamma = 0.1$, and valuation parameters $u_L^l = 1$, $u_L^h = 2$, $u_H^l = 4$, $u_H^h = 8$. Panel A plots market liquidity as a function of q for the steady state. Panel B also plots peak market liquidity for the two- and three-period cycles for $q \in [\underline{q}_1, \bar{q}_1]$. A solid line indicates a stable equilibrium. Computations for π_0 and π^{ss} are reported in [Internet Appendix IA.V](#). (Color figure can be viewed at [wileyonlinelibrary.com](#))

cycle without mixing, the peak-to-trough ratio $\frac{\mu_{H,2}^l}{\mu_{H,1}^l}$ decreases with γ , which explains why the existence condition is easier to satisfy when γ is small. When $\gamma \rightarrow 0$, however, equilibrium fluctuations also become smaller: the difference $\mu_{H,2}^l - \mu_{H,1}^l$ converges to zero when $\gamma \rightarrow 0$. Without liquidity shocks, the cycle degenerates to the no-trade steady state.

Corollary 1 suggests that longer cycles are more likely to arise when investors have less frequent rebalancing needs. Housing can be seen as a low- γ asset for which long booms are followed by quick busts. Financial assets such as corporate bonds should instead exhibit high-frequency fluctuations. Other predictions for these two markets are discussed in Section V.

A.1. Stable Equilibrium

Proposition 2 shows that a cycle coexists with a mixed-strategy steady state. If market quality is intermediate, no pure-strategy steady state is sustainable. A constantly high price would drive the effective supply of H assets to its lowest value γq by the quality effect. If $q \leq \bar{q}_1$, it is too low to sustain a high-price equilibrium. Conversely, the effective supply q with a constantly low price would be too high to rationalize this offer for $q \geq \underline{q}_1$. Hence, if $q \in [\underline{q}_1, \bar{q}_1]$, a steady-state equilibrium is possible only if buyers mix between offers with a constant probability. In the coexisting cycle, the quality effect instead creates fluctuations between offers over time.

Figure 3 plots market liquidity when existence condition (C_T) holds only for $T \leq 3$. Panel 3(a) focuses on the steady state. Panel 3(b) zooms on the parameter range $[\underline{q}_1, \bar{q}_1]$ over which cycles can coexist. Market liquidity is zero in a trough, so only peak market liquidity is represented. Panel 3(b) suggests that

when cycles coexist, the longer the cycle is, the higher peak liquidity is. Intuitively, the average trading probability over time is pinned down by the same underlying market quality. Hence, a longer trough must be associated with a higher level of peak trading.

The next result shows that when several equilibria coexist, the stability criterion selects the cycle when it exists.

COROLLARY 2: *When condition (C_1) holds, there exists at most one stable stationary equilibrium. For $q \leq \underline{q}_1$ ($q \geq \bar{q}_1$), the unique stable equilibrium is the illiquid (liquid) steady state. For $q \in (\underline{q}_1, \bar{q}_1)$, the unique stable equilibrium—if it exists—is the period T cycle for $q \in (\underline{q}_T, \bar{q}_T)$.*

The stability criterion selects the unique equilibrium without buyer mixing between offers. Intuitively, if buyers play a pure strategy in equilibrium, the equilibrium offer may still be strictly preferred after a small perturbation of the state variable μ_H^l . The proof shows how to construct a transition path following such a perturbation. Convergence can occur by the first period a high offer is made, if any, in the original stationary equilibrium. Intuitively, after a high offer, law of motion (4) shows that the effective supply is reset to its lowest level γq , which belongs to the state space of the stationary equilibrium. In the illiquid steady state, a high offer is never made in equilibrium, so convergence occurs in the limit, as illustrated by Panel 1(a).

When market quality q is intermediate, Corollary 2 justifies the focus on cycles in which buyers do not mix between offers. The results above also imply that there does not exist a stable stationary equilibrium for all values of q . For instance, when (C_T) holds for $T = 1$ only, there is no stable equilibrium when $q \in (\underline{q}_1, \bar{q}_1)$.²⁰

A.2. Volume and Prices in Cycles

I conclude this section by discussing the qualitative properties of the (unique) stable period T cycle when the existence condition (C_T) holds. As shown by Proposition 2, in a cycle, buyers make a high offer at the peak and a low offer in trough periods. Trading volume thus follows a simple pattern. During a trough, only lL asset owners sell for a constant volume of $\mathcal{V}_{\text{trough}}^{(T)} := \gamma(1 - q)$. At the peak, all asset owners except hH investors take the high offer and sell. Trading volume is then

$$\mathcal{V}_{\text{peak}}^{(T)} := 1 - q + \mu_{H,0}^{l,(T)} = 1 - q + [1 - (1 - \gamma)^T]q. \quad (18)$$

The volume spikes at the peak for two reasons. First, the effective supply of H assets is finally traded. As shown by the second term of (18), the longer the cycle is, the higher is this component of supply due to the longer accumulation

²⁰ When $\delta = 0$, Proposition 1 instead shows that there exists a unique stable equilibrium for all values of q . In this case, the existence condition (C_T) holds for any $T \geq 1$, and the intervals $[\underline{q}_T, \bar{q}_T]$ for $T \geq 2$ form a partition of the interval $[\underline{q}_1, \bar{q}_1]$.

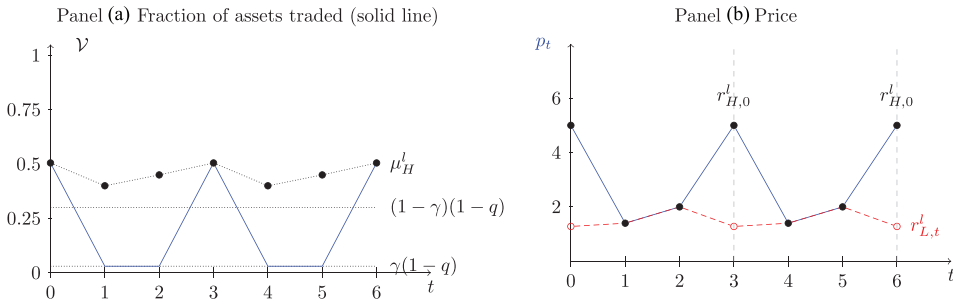


Figure 4. Volume and price in a three-period cycle. Parameters: market quality $q = 0.68$, discount factor $\delta = 0.2$, and valuation parameters $u_L^L = 1$, $u_H^L = 4$, $u_a^h = 2u_a^l$, $\gamma = 0.1$. Panel A plots the trading volume. The values reported on the right of the plot correspond to the difference relative to the next lower-dashed line. Panel B plots the asset price (solid blue line) over the three-period cycle. The reservation price of LH asset owners is the dashed red line. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jof.13106))

of selling pressure in an illiquid market. Hence, markets that are quieter for a longer period are more prone to a sudden increase in trading volume. The second extra component of supply compared to the trough comes from hL asset owners who opportunistically resell their asset at the peak. Figure 4 illustrates this pattern in three-period cycle.

In a cycle, volume dynamics are shaped by the quality effect. Price dynamics are driven instead by the resale effect. In a stable T -period cycle, the peak price is equal to the reservation price of LH investors, r_H^L . In period $t \in \{1, \dots, T-1\}$ of the trough, the price is given by the reservation price of LL investors, given by

$$r_{L,t}^{L,(T)} = \frac{1 - \delta^{T-t}}{1 - \delta} u_L^L + \delta^{T-t} r_H^L, \quad (19)$$

where the derivations are given in the proof of Prediction 2 below. The first component of the price is the discounted holding value flow until the peak; the second component is the discounted peak price. The closer the peak, the higher the L asset price due to the increasing option value to wait for the peak to sell at the pooling price r_H^L . Panel 4(b) illustrates this pattern in a three-period cycle.

In this weak resale effect case, the model makes strong predictions for the dynamics of asset prices. Cyclical fluctuations can arise in opaque markets with resale, even in the absence of shocks to asset fundamentals. Section V provides two applications of these results. First, cycles are related to excess price fluctuations in the corporate bond market. Second, I use the model to shed light on speculative price dynamics documented in the housing market and other asset classes. Before this discussion, the next section completes the equilibrium analysis by describing the strong resale effect case.

B. Strong Resale Effect

I now characterize stable stationary equilibria when condition (C_1) does not hold, that is, when the resale effect is strong. This analysis confirms the result in Section IV.A that cycles exist only when investors' discount factor is low. Instead, with a strong resale effect, steady-state multiplicity arises.

PROPOSITION 3: *If (C_1) does not hold, there is no cycle. The market quality thresholds in Proposition 2 satisfy $\bar{q}_1 < \underline{q}_1$ and*

- i) for $q \leq \bar{q}_1$, the unique stable steady state is illiquid ($\pi^{ss} = 0$),*
- ii) for $q \geq \underline{q}_1$, the unique stable steady state is liquid ($\pi^{ss} = 1$), and*
- iii) for $q \in (\bar{q}_1, \underline{q}_1)$, the liquid steady state and the illiquid steady state coexist and are the only stable equilibria.*

This result is similar to Proposition 2 in Chiu and Koepl (2016). The unique stable equilibrium is again the liquid (illiquid) steady state when market quality q is high (low). When q is intermediate, however, with a strong resale effect, equilibrium liquidity dynamics are different. First, there is no cycle: when the discount factor is high, liquidity today cannot be different from liquidity tomorrow. Second, for intermediate market quality q , the liquid steady state now coexists with the illiquid steady state.

Equilibrium multiplicity for intermediate values of market quality q is due to the resale effect. As explained above, if future buyers offer a high (low) price, current buyers are more likely to offer a high (low) price. When investors are very forward-looking, this intertemporal complementarity leads to equilibrium multiplicity. Formally, the threshold $\underline{q}_1$ below which an illiquid steady state exists is increasing with the discount factor δ and eventually exceeds $\bar{q}_1$ when δ is high enough. Intuitively, in an illiquid steady state, a buyer who would deviate to make a high offer would pay the present value of information rents. This cost of deviating to a high offer is thus increasing in the discount factor δ . Hence, an illiquid steady state can be sustained for higher values of q . When there is coexistence, the illiquid steady state is inefficient because of the lack of intertemporal coordination between buyers.

In the strong resale effect case, the quality effect does not generate fluctuations, but it still implies a negative relationship between the quality of assets for sale and market liquidity. Consider a value of market quality q in the interval $[\bar{q}_1, \underline{q}_1]$ for which steady states coexist. From law of motion (4), the effective supply of H assets is γq in the liquid steady state, below the effective supply q in the illiquid steady state, because $\gamma < 1$. High market liquidity implies that H assets flow to h valuation buyers, who do not resell unless they receive a liquidity shock. The pool of sellers is thus less favorable in the liquid steady state. Prices are mostly determined by the resale effect; however, a high offer is optimal despite the worse pool of assets if assets can be resold at a high price.

V. Empirical Implications

This section derives empirical predictions with applications for two asset classes motivating the analysis: corporate bonds and housing. As already argued, these two markets are reasonable testing grounds for the model because information asymmetry is plausible and trading is decentralized.

These predictions are derived from the novel weak-resale-effect case analyzed in Section IV.A in which the equilibrium is a cycle for intermediate values of market quality. In this case, with few ingredients, the model delivers a rich set of predictions. In Section V.A, I argue that the model provides a liquidity-based explanation for excess price fluctuations in corporate bonds. In Section V.B, I interpret cycles as repeated speculative episodes.

A. Excess Price Fluctuations

In an equilibrium cycle, prices fluctuate while asset fundamentals are constant. By construction, the share of high-quality assets in the market is fixed at q , but the composition of assets sold, and thus liquidity, change endogenously over time.

PREDICTION 1: *In opaque asset markets, prices fluctuate more than fundamentals due to time-varying liquidity.*

Prediction 1 offers a liquidity-based rationale for excess price fluctuations, similar to Daley and Green (2015). These fluctuations have been documented for corporate bonds and housing by Bao and Pan (2013) and Glaeser et al. (2014), respectively. An alternative explanation may be that these patterns are generated by the same discount rate fluctuations that explain excess volatility in stocks (see Shiller (1981)). However, this explanation is not supported by Kuvshinov (2020) who finds a weak correlation between the discount rates for equity, housing, and corporate bonds. Consistent with a liquidity-based explanation, Bai, Bali, and Wen (2019) show that liquidity is an important driver of bond prices, which is distinct from traditional stock factors. Their liquidity measure, as well as the bond illiquidity metric in Bao, Pan, and Wang (2011), exhibit frequent fluctuations, consistent with Prediction 1.

The cyclical nature of the equilibrium imposes strong restrictions on fluctuations: prices rise and then predictably fall. To translate this observation into a testable empirical implication, I define a measure of price reversal as

$$\gamma := -\text{Cov}(\Delta p_t, \Delta p_{t+1}), \quad (20)$$

with $\Delta p_t = p_t - p_{t+1}$ the first-order price change. To compute moments, a weight $\frac{1}{T}$ is assigned to each equilibrium state of the state space (see Definition 2). The price reversal measure defined in (20) is similar to the illiquidity proxy in Bao, Pan, and Wang (2011). These authors interpret a positive value of γ as evidence of liquidity-driven reversals in corporate bonds returns.

PREDICTION 2: *In a cycle, price changes exhibit reversal, that is, $\gamma > 0$.*

The proof of this result is in Appendix A. Although there is short-term momentum in the trough of a cycle of period $T \geq 3$, the unconditional price reversal measure γ is negative. This negative sign is driven by the strong reversal around the cycle peak: the last price increase in the trough Δp_T is the largest and the magnitude of the ensuing price drop Δp_1 is equal to the sum of all price increases in the trough. Prediction 2 is consistent with the empirical evidence in Bao, Pan, and Wang (2011) for corporate bonds.²¹ Because price reversals are driven by liquidity in the model, the result also corroborates their interpretation of γ as a proxy for illiquidity. In support of information asymmetry as a driver of illiquidity, Ivaschenko (2020) shows in a recent paper that price reversals are stronger for more opaque bonds.

B. Speculation

In this section, I interpret cycles as repeated speculative episodes, also called bubbles. This interpretation is motivated by the observation that the price of an L asset in a cycle trough is driven not by its fundamental value but by its resale value at the peak, as shown by equation (19). Harrison and Kreps (1978) define speculative behavior as a trader's willingness to pay more than the fundamental value of an asset because he can later resell it at a higher price. As the notion of fundamental value is ambiguous in this model, I compare the equilibrium price to a conservative benchmark for the value of the L asset, given by $\bar{p}_L := \frac{u_L^h}{1-\delta}$. This value is the discounted utility flow of an asset held by a high-valuation investor in every period.²²

PREDICTION 3: *In a cycle trough, the L asset price exceeds the fundamental value $\bar{p}_L$ if $\delta > \frac{u_L^h - u_L^l}{u_H^l - u_L^l}$.*

Prediction 3 follows from comparing $\bar{p}_L$ to the L asset price in the last trough period, using equation (19). In the limit $u_L^h \rightarrow u_L^l$, the condition in Prediction 3 is trivial, and thus, it can hold jointly with the existence condition for a cycle (C_T). Prediction 3 is intuitive: prices exceed fundamental values if the resale effect generating speculation is strong (high δ).

Equilibrium speculation is not a trivial result, even under asymmetry of information. Obviously, an L asset trades at a speculative price at the peak of the cycle because it is pooled with high-quality assets. However, Prediction 3 shows that speculation can happen during the trough, when only L assets are traded, that is, when the offer is separating. The speculative component reflects buyers' ability to later resell a lemon in a good pool at the peak. Such

²¹ Bali, Subrahmanyam, and Wen (2021) also provide evidence of reversals at long horizons. There is also momentum in corporate bonds return but the phenomenon seems to be confined to the small noninvestment-grade segment of the market (see Jostova et al. (2013)).

²² This price would obtain under symmetric information if sellers, instead of buyers, had the bargaining power. Under the current allocation of bargaining power, the symmetric information price is given by $\frac{u_L^h}{1-\delta} < \bar{p}_L$. With this benchmark, trough prices are always speculative. The result is also robust to using the buy-and-hold valuation as a benchmark.

speculation can arise because the trading history is opaque. This prediction is also made by Asriyan, Fuchs, and Green (2019) in a similar environment.

Liquidity cycles may thus contribute to speculative dynamics observed for several assets (see Cutler, Poterba, and Summers (1991)). Glaeser et al. (2014) identify these dynamics in the housing market and, more recently, Enoksen et al. (2020) show that cryptocurrencies exhibit recurring bubble episodes.

As in Section V.A, I show that the model not only provides a story for the empirical pattern of interest—in this case, price speculation—but also offers stark predictions supported by empirical evidence.

PREDICTION 4: *For a cycle of period $T \geq 3$, the rate of price change increases during the boom, that is, $\Delta p_{t+1} \geq \Delta p_t$, for $2 \leq t \leq T - 1$.*

The proof of Prediction 2 shows that the change in the asset price in the trough of a T -period cycle is given by $\Delta p_t = \delta^{T-t}(u_H^l - u_L^l)$. Prediction 4 follows immediately. In a cycle, the price increases when $T - t$ decreases due to the increasing resale value of the asset. This price change also increases because the difference in the expected resale value increases as the peak gets closer. In the model, a speculative episode is thus characterized by acceleration before the bubble bursts. Greenwood, Shleifer, and You (2019) show that acceleration is a typical feature of price rallies that end up in busts.

Finally, many accounts of bubbles describe the asset as a “hot potato” that speculators pass to the next investors. To formalize this narrative, I compute the expected holding time of the asset along the equilibrium cycle. The expected holding time in period t , denoted as EHT_t , is defined as the average expected time before an asset bought in period t is resold by its owner. The next result shows that this holding time decreases during a boom.

PREDICTION 5: *In a cycle of period T , the expected holding time of an asset is*

$$EHT_t = \begin{cases} \frac{\gamma^{-1}[1 - (1 - \gamma)^T] + qT}{1 - q + q[1 - (1 - \gamma)^T]} & \text{if } t = 0 \\ \gamma^{-1}[1 - (1 - \gamma)^{T-t}] & \text{if } t = 1, \dots, T - 1, \end{cases} \quad (21)$$

which is strictly decreasing in t for $t = 0, \dots, T - 1$.

The proof of Prediction 5 is in Appendix A. The intuition for the result is as follows. Consider first the trough during which only L assets are traded. At the peak, all L asset owners sell, and thus, the expected holding time of an asset bought in the previous period is equal to one. An investor who buys two periods before the peak must wait for the peak to resell unless he receives a liquidity shock. Hence, the expected holding time is longer. Extending the logic, the expected holding time in the trough increases with the time to the next peak. For an asset bought at the peak, another effect plays a role: at the peak, an asset bought can be of high quality and, even if the new owner then received a liquidity shock, she would only trade at the next peak.

This pattern of expected holding time is consistent with the empirical evidence in Zwick, Nathanson, and DeFusco (2020) that there was a dramatic

increase in speculative volume during the 2000/2005 housing boom. Following Bayer et al. (2020), they identify speculative episodes as times when the expected holding period of a house decreases. Prediction 5 is consistent with this characterization of speculation.

VI. Welfare

In this section, I derive the normative properties of stationary equilibria under a utilitarian criterion. The source of inefficiency is that some gains from trade for the H asset are not realized when the market is illiquid. As all IL investors always sell their asset, asset misallocation in period t can be defined as the fraction of lH investors who do not trade, that is,

$$Misalloc_t = \mu_{H,t}^l [1 - \pi_t]. \quad (22)$$

High market liquidity π_t in period t reduces misallocation. However, by virtue of the quality effect, the future supply of H assets $\mu_{H,t+1}^l$ decreases with π_t , potentially increasing future misallocation in period $t+1$. This effect is at play in cycles where periods of illiquidity follow a period of high liquidity. To simplify the comparison between the steady state and cycles, I make the following assumption.

ASSUMPTION 3: (C_T) holds only for $T = 2$, $q \in [\underline{q}_1, \bar{q}_1]$, and $u_L^h = u_L^l = u_L$.

The first part of Assumption 3 implies that only two stationary equilibria exist: a steady state in which buyers make a high offer with probability π^{ss} , and a two-period cycle with liquidity $\pi_0^{(2)} \in (0, 1]$ at the peak and $\pi_1^{(2)} = 0$ in the trough. Focusing on two-period cycles helps deliver the intuition about the welfare properties of cycles. The second part of Assumption 3 implies there is no gain from trading the L asset. This part of the assumption helps obtain closed-form solutions for equilibrium market liquidity, but has a limited impact on the welfare analysis because gains from trading the L asset are always realized.

To compare the steady state and the cycle, I compute the average misallocation in the cycle with equal weights for peak and trough periods. Using equation (22), the two-period cycle delivers higher welfare than the steady state if and only if

$$\frac{\mu_1^{(2)} + (1 - \pi_0^{(2)})\mu_0^{(2)}}{2} \leq (1 - \pi^{ss})\mu^{ss}, \quad (23)$$

where the left-hand side obtains because $\pi_1^{(2)} = 0$ in the trough of a cycle. If the composition of the asset pool were fixed, it would be enough to compare the average value $\frac{\pi_0^{(2)}}{2}$ of market liquidity in a cycle to market liquidity in the

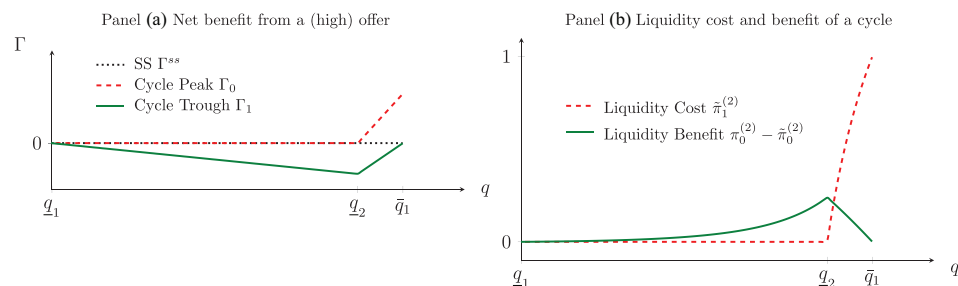


Figure 5. Intuition for welfare ranking. Parameters: discount factor $\delta = 0.5$, probability of a shock $\gamma = 0.1$, and valuation parameters $u_L = 1$, $u_H^l = 2$, $u_H^h = 4$. Panel A plots the net benefit from a high offer, given by (13). Panel B plots the illiquidity cost and the liquidity benefit in a cycle, defined in the text. Computations are reported in Internet Appendix IA.V. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com))

steady state π^{ss} . However, the composition of the pool itself depends on the path of market liquidity.²³ Still, it is possible to rank these equilibria.

PROPOSITION 4: *There exists a threshold for market quality $\hat{q} \in (\underline{q}_2, \bar{q}_1)$ such that the cycle delivers higher welfare if and only if $q \leq \hat{q}$.*

Proposition 4 states that for $q \in [\hat{q}, \bar{q}_1]$, the stable two-period cycle is less efficient than the unstable steady state. In Internet Appendix IA.IV, I show that a countercyclical policy with subsidies and taxes can then increase welfare. For market quality $q \in [\underline{q}_2, \hat{q}]$, the cycle is more efficient.

I now explain the welfare comparison in Proposition 4, focusing first on the region $[\underline{q}_1, \underline{q}_2]$. Recall that the market is liquid in period t if $\Gamma_t \geq 0$ with Γ_t the net benefit of a high offer. In the steady state, this condition holds as an equality. In every period, there is just enough liquidity such that, given the quality effect, a high offer is again weakly profitable the next period. In the cycle, liquidity is higher at the peak compared to the steady state, that is, $\pi_0^{(2)} > \pi^{ss}$, but this gain is not fully offset by lower liquidity in the trough. The welfare gain of a cycle over the steady state can thus be measured by the difference between equilibrium market liquidity at the peak, $\pi_0^{(2)}$, and the lower amount $\tilde{\pi}_0^{(2)}$ that would be necessary to also sustain liquidity in the trough. Panel 5(b) of Figure 5 plots this difference as a function of q . The hump-shaped pattern of this liquidity benefit reflects the U-shaped pattern of the (negative) net benefit from trading in the trough Γ_1 in Panel 5(a).

Due to this liquidity benefit, the cycle dominates the steady state when $q \in [\underline{q}_1, \underline{q}_2]$. When $q \in [\underline{q}_2, \bar{q}_1]$, however, there is a liquidity cost of the cycle relative to the steady state. In this region, buyers make a strictly positive profit at the cycle peak, that is, $\Gamma_0 > 0$, as illustrated in Panel 5(a), but this profit is earned at the expense of liquidity in the trough. To see this, define $\tilde{\pi}_1^{(2)} > 0$ as

²³ In the case $\delta = 0$, it is possible to show that this naive comparison is exact. In particular, when $q \in [\underline{q}_1, \underline{q}_2]$, market liquidity satisfies $\pi_0^{(2)} \geq 2\pi^{ss}$ and welfare is higher in the cycle.

the maximum amount of trading in the trough compatible with buyers trading at the peak.²⁴ Increasing trading in the trough from its equilibrium value $\pi_1^{(2)} = 0$ to $\tilde{\pi}_1^{(2)}$ has a clear benefit with no social cost at the peak. It would reduce the buyers' profit but the market would still be liquid, by definition of $\tilde{\pi}_1^{(2)}$. Hence, the variable $\tilde{\pi}_1^{(2)}$, plotted in Panel 5(b), measures the liquidity cost of a cycle compared to the steady state. This cost is strictly positive only in the region $[q_2, \bar{q}_1]$ in which buyers make a strictly positive profit. When the liquidity cost of a cycle exceeds the benefit described above, the steady state dominates.

The result that the cycle is less efficient when market quality q is close to the threshold $\bar{q}_1$ for a liquid steady state is not surprising. In the limit as q approaches $\bar{q}_1$, there is no steady-state misallocation because market liquidity π^{ss} converges toward one. In a cycle, however, every second period, a small mass γq of lH investors do not trade: misallocation persists even when q is very close to $\bar{q}_1$. Unlike in the steady state, an increase in market quality translates into a higher buyer's profit at the peak but average liquidity does not change.

Overall, the welfare analysis shows that although investors are risk-neutral, fluctuations matter from a welfare point of view. Welfare in the (stable) cyclical equilibrium can be higher than in the (unstable) steady-state equilibrium. This result arises because cyclical liquidity is high precisely when the effective supply of H assets is high, which reduces misallocation.

VII. Conclusion

I present a model to characterize the liquidity dynamics of a secondary market in which asset quality is heterogeneous and trade history is opaque. With few plausible ingredients, the model generates rich dynamics. When market quality is intermediate, endogenous liquidity cycles can emerge in the absence of changes to fundamentals. Cycles arise because of a quality effect whereby past trading intensity has a negative effect on current market liquidity. Price and volume dynamics are shown to be speculative due to a resale effect. These fluctuations can be efficient compared to a steady-state equilibrium with a constant level of trade. The model also shows that when investors are more patient, cycles disappear but steady states with different degrees of liquidity coexist. The findings are useful to rationalize empirical patterns for several asset classes, including corporate bonds.

The analysis leaves several avenues for future research. While the supply of assets is fixed and constant in my model, primary market issuance could react to secondary market liquidity. Moreover, incentives to produce high-quality assets may change along the cycle. Finally, it would be interesting to investigate how policies to increase transparency or efforts to migrate OTC trading toward centralized markets would affect the equilibrium in this environment.

²⁴ Such trading would generate losses for buyers by definition of the trough of a cycle.

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Appendix A: Proofs

A.1. Proof of Lemma 1

Let $(i, j) \in \{l, h\}^2$ with $j \neq i$ and $(a, a') \in \{L, H\}^2$ such that $u_a^i > u_{a'}^j$. The dynamic equations for reservation prices are given by

$$r_{a,t}^i = u_a^i + \delta \mathbb{E}^{i_+|i} [v_{a,t+1}^{i_+}] = u_a^i + \delta \mathbb{E}^{i_+|i} \left[\int_p \max \{r_{a,t+1}^{i_+}, p\} dF_{t+1}(p) \right], \quad (\text{A.1})$$

where $\mathbb{E}^{i_+|i}[\cdot]$ denotes the expectation operator taken over investor i 's private valuation i_+ next period. By assumption, if $i = l$, then $i_+ = l$ while if $i = h$, then $i_+ = h$ with probability $1 - \gamma$ and $i_+ = l$ otherwise. It is suboptimal for an investor to mimic the acceptance strategy of another type, so

$$r_{a,t}^i \geq u_a^i + \delta \mathbb{E}^{i_+|i} \left[F_{t+1}(r_{a',t+1}^{j_+}) r_{a,t+1}^{i_+} + \int_{p \geq r_{a',t+1}^{j_+}} p dF_{t+1}(p) \right]. \quad (\text{A.2})$$

The following inequalities thus obtain:

$$\begin{bmatrix} r_{L,t}^h - r_{L,t}^l \\ r_{H,t}^l - r_{L,t}^h \\ r_{H,t}^h - r_{H,t}^l \end{bmatrix} \geq \begin{bmatrix} u_L^h - u_L^l \\ u_H^l - u_L^h \\ u_H^h - u_H^l \end{bmatrix} + \delta N_{t+1} \begin{bmatrix} r_{L,t+1}^h - r_{L,t+1}^l \\ r_{H,t+1}^l - r_{L,t+1}^h \\ r_{H,t+1}^h - r_{H,t+1}^l \end{bmatrix}, \quad (\text{A.3})$$

$$\text{where } N_{t+1} = \begin{pmatrix} (1-\gamma)F_{t+1}(r_{L,t+1}^l) & 0 & 0 \\ \gamma F_{t+1}(r_{L,t+1}^l) & F_t(r_{L,t}^h) & 0 \\ 0 & 0 & (1-\gamma)F_{t+1}(r_{H,t+1}^l) \end{pmatrix}. \quad (\text{A.4})$$

All elements of N_{t+1} lie in $[0, 1]$. Iterating on (A.3) and using the transversality condition $\lim_{t \rightarrow \infty} \delta^t r_{a,t}^i = 0$, it follows that the left-hand side of (A.3) is positive because the first term on the right-hand side of (A.3) is positive. This concludes the proof.

A.2. Proof of Lemma 2

The first step shows that $p \in \cup_{i,a} r_{a,t}^i$. The second step shows that offers $r_{L,t}^h$ and $r_{H,t}^h$ are suboptimal.

Consider two consecutive types $ia \succ ja'$ ranked according to Assumption 1, where $(i, j) \in \{l, h\}^2$ and $(a, a') \in \{L, H\}^2$. Let $p \in (r_{a',t}^j, r_{a,t}^i)$. It immediately follows from equation (10) that the buyer's profit $v_{B,t}$ is higher with offer $r_{a',t}^j$ than with offer p .

For the second part of the proof, consider offer $r_{L,t}^h$ (the proof is similar for offer $r_{H,t}^h$). From equation (10), a buyer who offers $r_{L,t}^l$ instead of $r_{L,t}^h$ increases

his profit by

$$\mu_{L,t}^l \left(r_{L,t}^h - r_{L,t}^l \right) \geq 0. \quad (\text{A.5})$$

Hence, it is profitable to reduce the offer to $r_{L,t}^l$. This concludes the proof.

A.3. Proof of Lemma 3

The following relationships can be derived from the proof of Lemma 1:

$$r_{H,t}^l = u_H^l + \delta v_{H,t+1}^l = u_H^l + \delta r_{H,t+1}^l, \quad (\text{A.6a})$$

$$r_{H,t}^h = u_H^h + \delta \left\{ (1 - \gamma) v_{H,t+1}^h + \gamma v_{H,t+1}^l \right\} = \delta \left\{ (1 - \gamma) r_{H,t+1}^h + \gamma r_{H,t+1}^l \right\}, \quad (\text{A.6b})$$

where in each line, the second equality follows from equation (9) and Lemma 1, which implies that H asset owners weakly prefer to keep their asset. Equation (14a) follows directly from iterating the first equation and using the transversality condition $\lim_{t \rightarrow \infty} \delta^t r_{a,t}^i = 0$. Equation (14b) follows from the first result using the second equation with $r_{H,t}^h = r_{H,t+1}^h = r_H^h$. This concludes the proof.

A.4. Proof of Proposition 2

The proof is in three parts. Section A.4.1 proves the results for steady states. Section A.4.4 proves the results for cycles. Section A.4.5 derives results specific to Proposition 1.

A.4.1. Steady States

The proof for steady states is by construction. Conjecture an equilibrium with a constant degree of liquidity $\pi^{ss} \in [0, 1]$. Derive the endogenous variables as functions of π^{ss} (using π^{ss} as an argument with a slight abuse of notation). Finally, check that the value of π^{ss} is compatible with the optimality condition (12) given the value of the endogenous variables derived in the previous step.

From law of motion (4), the steady state value of μ_H^l is given by

$$\mu_H^l(\pi^{ss}) = \frac{\gamma q}{1 - (1 - \gamma)(1 - \pi^{ss})}. \quad (\text{A.7})$$

For the steady-state value of the second term of expression (13), use equations (15a) and (15b) to obtain

$$r_L^h(\pi^{ss}) = u_L^h + \delta \left\{ (1 - \pi^{ss}) \left[(1 - \gamma) r_L^h(\pi^{ss}) + \gamma r_L^l(\pi^{ss}) \right] + \pi^{ss} r_H^l(\pi^{ss}) \right\}, \quad (\text{A.8})$$

$$r_L^l(\pi^{ss}) = u_L^l + \delta \left\{ (1 - \pi^{ss}) r_L^l(\pi^{ss}) + \pi^{ss} r_H^l(\pi^{ss}) \right\}. \quad (\text{A.9})$$

Combining these equations yields

$$r_H^l(\pi^{ss}) - r_L^l(\pi^{ss}) = \frac{u_H^l - u_L^l}{1 - \delta(1 - \pi^{ss})}, \quad (\text{A.10})$$

$$r_H^h(\pi^{ss}) - r_L^h(\pi^{ss}) = \frac{u_H^h - u_L^h}{1 - \delta(1 - \gamma)(1 - \pi^{ss})}. \quad (\text{A.11})$$

These two equations imply that steady-state information rents are given by

$$r_H^l - (1 - \gamma)r_L^h(\pi^{ss}) - \gamma r_L^l(\pi^{ss}) = \frac{u_H^l - u_L^l}{1 - \delta(1 - \pi^{ss})} - (1 - \gamma) \frac{u_L^h - u_L^l}{1 - \delta(1 - \gamma)(1 - \pi^{ss})}. \quad (\text{A.12})$$

Illiquid Steady State $\pi^{ss} = 0$

Given the optimality condition (13), such a steady state exists if $\Gamma(0) < 0$, where

$$\Gamma(0) = q(r_H^h - r_H^l) - (1 - q) \left[\frac{u_H^l - u_L^l}{1 - \delta} - (1 - \gamma) \frac{u_L^h - u_L^l}{1 - \delta(1 - \gamma)} \right].$$

This condition is equivalent to $q \leq \underline{q}_1$, where

$$\underline{q}_1 := \frac{\frac{u_H^l - u_L^l}{1 - \delta} - (1 - \gamma) \frac{u_L^h - u_L^l}{1 - \delta(1 - \gamma)}}{r_H^h - r_H^l + \frac{u_H^l - u_L^l}{1 - \delta} - (1 - \gamma) \frac{u_L^h - u_L^l}{1 - \delta(1 - \gamma)}}. \quad (\text{A.13})$$

A.4.2. Liquid Steady State $\pi^{ss} = 1$

Using optimality condition (13), such a steady state exists if $\Gamma(1) > 0$, where

$$\Gamma(1) = \gamma q(r_H^h - r_H^l) - (1 - q) \left[(u_H^l - u_L^l) - (1 - \gamma)(u_L^h - u_L^l) \right].$$

The condition is equivalent to $q \geq \bar{q}_1$ where

$$\bar{q}_1 := \frac{(u_H^l - u_L^l) - (1 - \gamma)(u_L^h - u_L^l)}{\gamma(r_H^h - r_H^l) + (u_H^l - u_L^l) - (1 - \gamma)(u_L^h - u_L^l)}. \quad (\text{A.14})$$

No Multiplicity

The liquid and illiquid steady states do not coexist if $\bar{q}_1 > \underline{q}_1$. Using equations (A.13) and (A.14), this condition is equivalent to

$$(u_H^l - u_L^l) - (1 - \gamma)(u_L^h - u_L^l) \geq \gamma \frac{u_H^l - u_L^l}{1 - \delta} - \gamma(1 - \gamma) \frac{u_L^h - u_L^l}{1 - \delta(1 - \gamma)}, \quad (\text{A.15a})$$

$$\frac{1 - \delta - \gamma}{1 - \delta} (u_H^l - u_L^l) \geq \frac{(1 - \delta)(1 - \gamma)^2}{1 - \delta(1 - \gamma)} (u_L^h - u_L^l). \quad (\text{A.15b})$$

The last line is equivalent to condition (C₁) in the main text.

A.4.3. Mixed-Strategy Steady State

The last step is to show that a mixed-strategy steady state with $\pi^{ss} \in (0, 1)$ exists when $q \in (q_1, \bar{q}_1)$. From condition (12), a mixed-strategy steady state exists if there exists $\pi^{ss} \in (0, 1)$ such that $\Gamma(\pi^{ss}) = 0$, where

$$\Gamma(\pi^{ss}) = \frac{\gamma q}{1 - (1 - \gamma)(1 - \pi^{ss})}(r_H^h - r_H^l) - (1 - q) \left[\frac{u_H^l - u_L^l}{1 - \delta(1 - \pi^{ss})} - (1 - \gamma) \frac{u_L^h - u_L^l}{1 - \delta(1 - \gamma)(1 - \pi^{ss})} \right]. \quad (\text{A.16})$$

The results above that the liquid steady state or the illiquid steady state do not exist when $q \in (q_1, \bar{q}_1)$ implies that $\Gamma(0) > 0$ and $\Gamma(1) < 0$. Because Γ is a continuous function of π^{ss} , by the intermediate value theorem, there exists $\pi^{ss} \in (0, 1)$ such that $\Gamma(\pi^{ss}) = 0$. The equation above further shows that π^{ss} is the solution to a second-order polynomial. Hence, as $\Gamma(0) < 0$ and $\Gamma(1) > 0$, there cannot exist another solution $\hat{\pi} \in (0, 1) \setminus \{\pi^{ss}\}$ of $\Gamma = 0$ as otherwise the sign of the derivative of Γ would change at least twice on $(0, 1)$. This concludes the first part of the proof.

A.4.4. Cycles

This part of the proof is divided into four steps. The first step fully characterizes the cyclical equilibria in which buyers do not randomize. The second step shows that when (C_T) holds, there also exists a cyclical equilibrium of period T over which buyers randomize at the peak. The proof that there is no other cyclical equilibrium in which buyers randomize is in [Internet Appendix IA.VI](#) for the case $T = 2$. Finally, I show that the existence condition for a T -period cycle can be written as (C_T) and, that is, satisfies the claimed properties.

Step 1: Pure strategy cycles.

LEMMA A.1: *In a cycle without mixing by buyers, if $\pi_t = 1$, then $\pi_{t+1} = 0$.*

PROOF: The proof is by contradiction. Suppose that there exists a date t such that $\pi_t = 1$ and $\pi_{t+1} = 1$. Law of motion (4) implies that $\mu_{H,t+1}^l = \gamma q$. This is the lower bound for μ_H^l , so $\mu_{H,t-1}^l \geq \mu_{H,t+1}^l$. Because $\pi_t = 1$, the information rents in period $t - 1$ are equal to their minimum value

$$r_{H,t-1}^l - (1 - \gamma)r_{L,t-1}^h - \gamma r_{L,t-1}^l = u_H^l - (1 - \gamma)u_L^h - \gamma u_L^l. \quad (\text{A.17})$$

Using equation (13), it follows that $\Gamma_{t-1} > \Gamma_{t+1} > 0$, where the second inequality is by assumption. This shows that $\pi_{t-1} = 1$ but also that $\pi_{t-k} = 1$ for all integers k by iteration. The only stationary equilibrium with this property is a liquid steady state in which buyers always make the high offer. This leads to a contradiction. $\square$

Lemma A.1 establishes that a cycle in which buyers play pure strategies has one peak period and $T - 1$ trough periods. The existence conditions for such an equilibrium are now established. Let $t = 0$ or $t = T$ depending on the context be the period in which buyers make a high offer and $t = 1, \dots, T - 1$ be the periods in which the buyers make a low offer. By law of motion (4),

$$\mu_{H,t}^l = [1 - (1 - \gamma)^t]q, \quad \text{for } t = 1, \dots, T. \quad (\text{A.18})$$

From equations (15a) and (15b), we have

$$r_{H,t}^l - r_{L,t}^l = u_H^l - u_L^l + \delta(1 - \pi_{t+1})(r_{H,t+1}^l - r_{L,t+1}^l), \quad (\text{A.19a})$$

$$r_{H,t}^l - r_{L,t}^l = u_L^h - u_L^l + \delta(1 - \gamma)(1 - \pi_{t+1})(r_{L,t+1}^h - r_{L,t+1}^l). \quad (\text{A.19b})$$

With $\pi_0 = 1$ and $\pi_t = 0$ for $t \in [1, T - 1]$, for $t = 0, \dots, T - 1$, we then have

$$r_{H,t}^l - r_{L,t}^l = \frac{1 - \delta^{T-t}}{1 - \delta}(u_H^l - u_L^l), \quad (\text{A.20a})$$

$$r_{H,t}^l - r_{L,t}^l = \frac{1 - [\delta(1 - \gamma)]^{T-t}}{1 - \delta(1 - \gamma)}(u_L^h - u_L^l). \quad (\text{A.20b})$$

A low offer is indeed optimal in period $t = 1, \dots, T - 1$ if $\Gamma_t < 0$, with

$$\begin{aligned} \Gamma_t = & [1 - (1 - \gamma)^t]q(r_H^h - r_H^l) - (1 - q)\frac{1 - \delta^{T-t}}{1 - \delta}(u_H^l - u_L^l) \\ & + (1 - q)(1 - \gamma)\frac{1 - [\delta(1 - \gamma)]^{T-t}}{1 - \delta(1 - \gamma)}(u_L^h - u_L^l). \end{aligned} \quad (\text{A.21})$$

The second (third) term of (A.21) is higher (lower) for $t = T - 1$ than for any $t \in [1, T - 2]$. This implies that if a low offer is optimal in period $T - 1$, it is also optimal for all periods $t = 1, \dots, T - 2$. Hence, two conditions must be satisfied for the conjectured sequence to be an equilibrium: $\Gamma_0 > 0$ and $\Gamma_{T-1} < 0$. Tedious but straightforward computations show that the first condition is equivalent to $q \geq \underline{q}_T$ where

$$\underline{q}_T := \frac{\frac{1 - \delta^T}{1 - \delta}(u_H^l - u_L^l) - (1 - \gamma)\frac{1 - [\delta(1 - \gamma)]^T}{1 - \delta(1 - \gamma)}(u_L^h - u_L^l)}{[1 - (1 - \gamma)^T](r_H^h - r_H^l) + \frac{1 - \delta^T}{1 - \delta}(u_H^l - u_L^l) - (1 - \gamma)\frac{1 - [\delta(1 - \gamma)]^T}{1 - \delta(1 - \gamma)}(u_L^h - u_L^l)}. \quad (\text{A.22})$$

The second condition is equivalent to $q \leq \bar{q}_T$, where

$$\bar{q}_T := \frac{u_H^l - (1 - \gamma)u_L^h - \gamma u_L^l}{[1 - (1 - \gamma)^{T-1}](r_H^h - r_H^l) + u_H^l - (1 - \gamma)u_L^h - \gamma u_L^l}. \quad (\text{A.23})$$

Such a cycle exists for $q \in [\underline{q}_T, \bar{q}_T]$ if $\underline{q}_T \leq \bar{q}_T$, that is, if

$$\frac{1 - (1 - \gamma)^T - [1 - (1 - \gamma)^{T-1}] \frac{1 - \delta^T}{1 - \delta}}{1 - (1 - \gamma)^T - [1 - (1 - \gamma)^{T-1}] \frac{1 - [\delta(1 - \gamma)]^T}{1 - \delta(1 - \gamma)}} \geq (1 - \gamma) \frac{u_L^h - u_L^l}{u_H^l - u_L^l}. \quad (\text{A.24})$$

In Step 3, I show that equation (A.24) can be expressed as (C_T) .

Step 2: Mixed-strategy cycles.

The same strategy is used to derive the existence condition of cycles with randomization at the peak. Let $\pi_0 \in (0, 1)$ be the (conjectured) probability of a high offer at the peak. In all other periods $t \in [1, T - 1]$, the conjecture is $\pi_t = 0$. Given this conjectured sequence $\{\pi_t\}_{t \in [0, T-1]}$, equilibrium existence requires that (i) there exists a value $\pi_0 \in (0, 1)$ such that $\Gamma_0 = 0$, where Γ_0 is the net gain from a high offer at the peak, and (ii) for this value of π_0 , $\Gamma_t < 0$ for all values of $t \in [1, T - 1]$.

The proof of point (i) relies on the properties of pure-strategy equilibria. By definition of $\underline{q}_T$, $\Gamma_0(1) \geq 0$ if and only if $q \geq \underline{q}_T$. By definition of an illiquid steady state, $\Gamma_0(0) \geq 0$ if and only if $q \geq \underline{q}_1$, where $\underline{q}_1 < \underline{q}_T$ for $T \geq 2$. The intermediate value theorem then implies that there exists $\pi_0 \in (0, 1)$ such that $\Gamma_0(\pi_0) = 0$ if and only if $q \in (\underline{q}_1, \underline{q}_T)$.

The argument of Lemma A.1 that a low offer is optimal in period $t \in [1, T - 1]$ if it is optimal in period $T - 1$ still holds. To prove point (ii), derive the value of the endogenous variables as a function of π_0 both at the peak and in period $T - 1$. Using the law of motion, we have

$$\mu_{H,0}^l = \frac{1 - (1 - \gamma)^T}{1 - (1 - \gamma)^T(1 - \pi_0)} q, \quad \mu_{H,T-1}^l = \mu_{H,0}^l - \frac{\pi_0(1 - \gamma)^{T-1}}{1 - (1 - \gamma)^T(1 - \pi_0)} \gamma q. \quad (\text{A.25})$$

For the reservation prices, we have

$$r_{H,0}^l - r_{L,0}^l = \frac{1 - \delta^T}{1 - \delta^T(1 - \pi_0)} \frac{u_H^l - u_L^l}{1 - \delta}, \quad (\text{A.26a})$$

$$r_{H,T-1}^l - r_{L,T-1}^l = (1 - \pi_0) [r_{H,0}^l - r_{L,0}^l] + \frac{\pi_0}{1 - \delta^T(1 - \pi_0)} (u_H^l - u_L^l), \quad (\text{A.26b})$$

and

$$r_{L,0}^h - r_{L,0}^l = \frac{1 - [\delta(1 - \gamma)]^T}{1 - [\delta(1 - \gamma)]^T(1 - \pi_0)} \frac{u_L^h - u_L^l}{1 - [\delta(1 - \gamma)]}, \quad (\text{A.27})$$

$$r_{L,T-1}^h - r_{L,T-1}^l = (1 - \pi_0) [r_{L,1}^h - r_{L,1}^l] + \frac{\pi_0}{1 - [\delta(1 - \gamma)]^T(1 - \pi_0)} (u_L^h - u_L^l). \quad (\text{A.28})$$

The condition that buyers mix in period 0, that is, $\Gamma_0 = 0$ can be written as

$$0 = \frac{1 - (1 - \gamma)^T}{1 - (1 - \gamma)^T(1 - \pi_0)} q (r_H^h - r_H^l) - (1 - q) \frac{1 - \delta^T}{1 - \delta^T(1 - \pi_0)} \frac{u_H^l - u_L^l}{1 - \delta}$$

$$+ (1-q)(1-\gamma) \frac{1 - [\delta(1-\gamma)]^T}{1 - [\delta(1-\gamma)]^T(1-\pi_0)} \frac{u_L^h - u_L^l}{1 - [\delta(1-\gamma)]}. \quad (\text{A.29})$$

The condition that $\Gamma_{T-1} < 0$ is equivalent to $\Gamma_0 - \Gamma_{T-1} > 0$, which writes

$$0 < \frac{(1-\gamma)^{T-1}}{1 - (1-\gamma)^T(1-\pi_0)} \gamma q - (1-q) \frac{\frac{1-\delta^T}{1-\delta} - 1}{1 - \delta^T(1-\pi_0)} (u_L^h - u_L^l) \\ + (1-q)(1-\gamma) \frac{\frac{1-[\delta(1-\gamma)]^T}{1-\delta(1-\gamma)} - 1}{1 - [\delta(1-\gamma)]^T(1-\pi_0)} (u_L^h - u_L^l). \quad (\text{A.30})$$

Again using the condition $\Gamma_0 = 0$ to substitute for the first term, the condition $\Gamma_{T-1} < 0$ is equivalent to the inequality

$$0 < \left[(1-\gamma)^{T-1} \frac{1-\delta^T}{1-\delta} - \frac{1-(1-\gamma)^T}{\gamma} \left\{ \frac{1-\delta^T}{1-\delta} - 1 \right\} \right] (u_L^h - u_L^l) \\ - \left[(1-\gamma)^{T-1} \frac{1-[\delta(1-\gamma)]^T}{1-\delta(1-\gamma)} - \frac{1-(1-\gamma)^T}{\gamma} \left\{ \frac{1-[\delta(1-\gamma)]^T}{1-\delta(1-\gamma)} - 1 \right\} \right] (1-\gamma)(u_L^h - u_L^l). \quad (\text{A.31})$$

Quick inspection shows that the condition above is equivalent to (A.24). This concludes Part 2 of the proof.

Step 3: Condition (C_T).

It remains to show that existence condition (A.24) for a T -period cycle can be expressed as (C_T). Rewrite (A.24) as $F(\delta, \gamma, T) \geq 0$ and let $R_u := \frac{u_L^h - u_L^l}{u_H - u_L}$. The claims are that the derivatives F_T and F_δ are negative.

Proof that $F_T \leq 0$: Equation (A.24) implies that $(1-\gamma)R_u \leq 1$. To see this, rewrite the left-hand side of (A.24) as

$$LHS = \frac{1 - \frac{1-(1-\gamma)^{T-1}}{1-(1-\gamma)^T} \frac{1-\delta^T}{1-\delta}}{1 - \frac{1-(1-\gamma)^{T-1}}{1-(1-\gamma)^T} \frac{1-[\delta(1-\gamma)]^T}{1-\delta(1-\gamma)}}. \quad (\text{A.32})$$

Because $x \mapsto \frac{1-x^T}{1-x}$ is an increasing function, it follows that $\frac{1-\delta^T}{1-\delta} \geq \frac{1-[\delta(1-\gamma)]^T}{1-\delta(1-\gamma)}$, which implies that $LHS \leq 1$. Hence, $(1-\gamma)R_u \leq 1$.

Now define $g : T \mapsto \frac{1-(1-\gamma)^{T-1}}{1-(1-\gamma)^T}$. Rewrite

$$g(T) = \left(1 + \gamma \frac{(1-\gamma)^{T-1}}{1-(1-\gamma)^{T-1}} \right)^{-1}. \quad (\text{A.33})$$

The term between parentheses increases with $(1-\gamma)^{T-1}$, which means that it decreases with T . This implies that $g'(T) > 0$. Hence, $F_T \leq 0$ if and only if

$$0 \geq -g'(T) \frac{1-\delta^T}{1-\delta} - g(T) \frac{\delta^T \log(\frac{1}{\delta})}{1-\delta}$$

$$+ (1 - \gamma)R_u \left[g'(T) \frac{1 - [\delta(1 - \gamma)]^T}{1 - \delta(1 - \gamma)} - g(T) \frac{(\delta(1 - \gamma))^T \log\left(\frac{1}{\delta(1 - \gamma)}\right)}{1 - \delta(1 - \gamma)} \right] \quad (\text{A.34a})$$

$$0 \geq -g'(T) \left[\frac{1 - \delta^T}{1 - \delta} - (1 - \gamma)R_u \frac{1 - [\delta(1 - \gamma)]^T}{1 - \delta(1 - \gamma)} \right] \\ - g(T) \left[\frac{\delta^T \log\left(\frac{1}{\delta}\right)}{1 - \delta} - (1 - \gamma)R_u \frac{(\delta(1 - \gamma))^T \log\left(\frac{1}{\delta(1 - \gamma)}\right)}{1 - \delta(1 - \gamma)} \right]. \quad (\text{A.34b})$$

Because $x \mapsto \frac{1-x^T}{1-x}$ is increasing and $(1 - \gamma)R_u \leq 1$, the first term in brackets is positive. Hence, with $g'(T) \geq 0$, it is enough to show that the second term in brackets is also positive. Again, using the fact that $(1 - \gamma)R_u \leq 1$, this is true if the mapping $k : x \mapsto \frac{x^T \log(\frac{1}{x})}{1-x}$ is increasing in x . Hence,

$$k'(x) = \frac{(Tx^{T-1} \log(\frac{1}{x}) - x^{T-1})(1-x) + x^T \log(\frac{1}{x})}{(1-x)^2} \quad (\text{A.35a})$$

$$= \frac{x^{T-1}}{(1-x)^2} \left[\left[T \log\left(\frac{1}{x}\right) - 1 \right] (1-x) + x \log\left(\frac{1}{x}\right) \right] \quad (\text{A.35b})$$

$$\geq \frac{x^{T-1}}{(1-x)^2} \left[\log\left(\frac{1}{x}\right) - (1-x) \right]. \quad (\text{A.35c})$$

The term in brackets is a decreasing function of x and is equal to zero for $x = 1$. This proves that $k'(x) \geq 0$, and, in turn, that $F_T \leq 0$.

Proof that $F_\delta \leq 0$: $F_\delta \leq 0$ if and only if

$$0 \geq -\frac{1 - \delta^T - T\delta^{T-1}(1 - \delta)}{(1 - \delta)^2} \\ + (1 - \gamma)R_u \frac{1 - [\delta(1 - \gamma)]^T - T[\delta(1 - \gamma)]^{T-1}(1 - [\delta(1 - \gamma)])}{(1 - \delta(1 - \gamma))^2}. \quad (\text{A.36})$$

Using the previous result that $F_T \leq 0$, it follows that (A.24) holds for $T \geq 2$ if it holds for $T = 2$. Condition (A.24) for $T = 2$ implies that

$$(1 - \gamma)R_u \leq \frac{1 - \delta - \gamma}{(1 - \delta)(1 - \gamma)}. \quad (\text{A.37})$$

Hence, the desired condition is satisfied if

$$0 \geq -\frac{1 - \delta^T - T\delta^{T-1}(1 - \delta)}{1 - \delta} \\ + \frac{1 - \delta - \gamma}{(1 - \gamma)} \frac{1 - [\delta(1 - \gamma)]^T - T[\delta(1 - \gamma)]^{T-1}(1 - [\delta(1 - \gamma)])}{(1 - \delta(1 - \gamma))^2} \quad (\text{A.38a})$$

$$\begin{aligned} \Leftarrow 0 \geq & -\frac{1 - \delta^T - T\delta^{T-1}(1 - \delta)}{(1 - \delta)^2} \\ & + \frac{1 - [\delta(1 - \gamma)]^T - T[\delta(1 - \gamma)]^{T-1}(1 - [\delta(1 - \gamma)])}{(1 - \delta(1 - \gamma))^2}, \quad (\text{A.38b}) \end{aligned}$$

where the implication follows from inequality $\frac{1 - \delta - \gamma}{(1 - \gamma)} \leq 1 - \delta$. To conclude, it remains to show that the mapping $l : x \mapsto \frac{1 - x^T - Tx^{T-1}(1 - x)}{(1 - x)^2}$ increases with x :

$$l(x) = \frac{\sum_{\tau=0}^{T-1} x^\tau - Tx^{T-1}}{1 - x} = \frac{\sum_{\tau=0}^{T-1} (x^\tau - x^{T-1})}{1 - x} = \frac{\sum_{\tau=0}^{T-1} x^\tau (1 - x^{T-1-\tau})}{1 - x} \quad (\text{A.39a})$$

$$= \frac{(1 - x) \sum_{\tau=0}^{T-1} x^\tau \sum_{i=0}^{T-1-\tau} x^i}{1 - x}. \quad (\text{A.39b})$$

Hence, l is increasing in x that implies that the desired condition holds.

A.4.5. Proof of Proposition 1

In this section, the additional results for the case $\delta = 0$ are proven. Given the results derived in the second part of the proof, it is enough to show that when $\delta = 0$, condition (C_T) holds for any T , $\bar{q}_{T+1} = \underline{q}_T$ for $T \geq 2$ and $\lim_{T \rightarrow \infty} \underline{q}_T = \underline{q}_1$. For the first result, note that the left-hand side of (C_T) converges to one when $\delta \rightarrow 0$, while the right-hand side converges to $(1 - \gamma) \frac{u_H^h - u_L^l}{u_H^h - u_L^l} < 1$. The second and third results follow directly from inspection of equations (A.22) and (A.23) when setting $\delta = 0$.

A.5. Proof of Corollary 1

As shown in the proof of Proposition 2, the existence condition for a T -period cycle is given by (A.24). When $u_L^h = u_L^l$, this condition becomes

$$\frac{1 - \delta^T}{1 - \delta} \leq \frac{1 - (1 - \gamma)^T}{1 - (1 - \gamma)^{T-1}} = 1 + \gamma \frac{(1 - \gamma)^{T-1}}{1 - (1 - \gamma)^{T-1}}. \quad (\text{A.40})$$

The left-hand side of (A.40) is an increasing function of $(1 - \gamma)^{T-1}$, and thus, is decreasing with γ . This proves that (C_T) is tightened as γ increases.

A.6. Proof of Corollary 2

The proof relies on two intermediate results. The first is that all equilibria in which buyers play only pure strategies are stable. This result shows that the equilibria mentioned in Corollary 2 are indeed stable. The second result rules out other equilibria in which buyers randomize between offers.

Step 1: Pure Strategy Equilibria Are Stable.

Consider a period, denoted as 0, of the stationary equilibrium and a deviation $\hat{\mu}_{H,0}^l$ from the original value $\mu_{H,0}^l$ such that $|\hat{\mu}_{H,0}^l - \mu_{H,0}^l| < \epsilon$. The objective is to show convergence to the initial stationary equilibrium with a sequence $\{\hat{\pi}_t\}_{t \geq 0} = \{\pi_t\}_{t \geq 0}$ identical to that of the unperturbed stationary equilibrium. Given this conjecture, the reservation prices are unchanged along the perturbed path. The conjectured equilibrium sequence $\{\pi_t\}_{t \geq 0}$ together with the law of motion and the initial value $\hat{\mu}_{H,0}^l$ determine a sequence $\{\hat{\mu}_{H,t}^l\}_{t \geq 0}$. The conjectured path is an equilibrium path if $\text{sgn}(\hat{\Gamma}_t) = \text{sgn}(\Gamma_t)$ for all $t \geq 0$, where $\hat{\Gamma}_t$ is the net benefit from a high offer in period t in the perturbed equilibrium. This path converges toward the stationary equilibrium if the long-run distribution of μ_H^l is the same as in the stationary equilibrium.

Let $t_1 \geq 0$ the first value of t (if any) such that $\pi_t = 1$ in the original stationary equilibrium. If $\hat{\Gamma}_{t_1} > 1$, then $\hat{p}_{i_{t_1}} = 1$ is a best response. This implies that $\hat{\mu}_{H,t_1+1}^l = \gamma q = \mu_{H,t_1+1}^l$. Hence, the conjectured equilibrium path from period t_1 onward is an equilibrium because it coincides with the stationary equilibrium. The last step is to prove that $\hat{\Gamma}_{t_1} > 0$ and $\hat{\Gamma}_t < 0$ for all $t \in [t_0, t_1 - 1]$ if ϵ is small enough. Given the conjecture that $\hat{\pi}_t = 0$ for all $t \in [t_0, t_1 - 1]$, the perturbed state variable is equal to

$$\hat{\mu}_{H,t}^l = q[1 - (1 - \gamma)^{t-t_0}] + (1 - \gamma)^{t-t_0} \hat{\mu}_{H,0}^l, \quad (\text{A.41a})$$

$$\hat{\mu}_{H,t}^l = \mu_{H,t}^l + (1 - \gamma)^{t-t_0} [\hat{\mu}_{H,0}^l - \mu_{H,0}^l]. \quad (\text{A.41b})$$

If $|\hat{\mu}_{H,0}^l - \mu_{H,0}^l| < \epsilon$, then $|\hat{\mu}_{H,t}^l - \mu_{H,t}^l| < \epsilon$ for all $t \in [t_0, t_1 - 1]$. Hence,

$$|\hat{\Gamma}_t - \Gamma_t| \leq \epsilon(r_H^h - r_H^l). \quad (\text{A.42})$$

The stationary equilibrium has buyers playing pure strategies so $\Gamma_t < 0$ for all $t \in [t_0, t_1 - 1]$. Hence, for ϵ small enough, $\hat{\Gamma}_t < 0$ and the conjectured sequence $\{\hat{\pi}_t\}_{t \in [t_0, t_1 - 1]} = \mathbf{0}$ constitutes an equilibrium path.

Finally, suppose that there is no t_1 such that $\pi_{t_1} = 1$ in the original equilibrium (illiquid steady state). In this case, because

$$|\hat{\mu}_{H,t}^l - \mu_{H,t}^l| \leq \epsilon(1 - \gamma)^t, \quad (\text{A.43})$$

the conjectured equilibrium with $\{\hat{\pi}_t\}_{t \geq 0} = \mathbf{0}$ is indeed an equilibrium path for ϵ small enough and $\lim_{t \rightarrow \infty} \hat{\mu}_{H,t}^l = \gamma q$. This proves convergence.

Step 2: Instability of Equilibria with Mixed Strategies.

CLAIM A.1: Let $\tau \geq 1$, and let a continuation equilibrium from period τ be given such that $\hat{\mu}_{H,\tau}^l \in (\gamma q, q)$ and $\hat{\mu}_{H,\tau}^l \neq \gamma(2 - \gamma)q$. An equilibrium path from $t = 0$ compatible with the continuation equilibrium defines a unique value $\hat{\mu}_{H,0}^l$.

PROOF (for $\tau = 1$): By law of motion (4), $\hat{\mu}_{H,0}^l$ and $\hat{\pi}_0$ must satisfy

$$\hat{\mu}_{H,1}^l = \gamma q + (1 - \gamma)(1 - \hat{\pi}_0)\hat{\mu}_{H,0}^l. \quad (\text{A.44})$$

If $\hat{\pi}_0 \in \{0, 1\}$, the result follows from the assumption that $\hat{\mu}_{H,1}^l > \gamma q$. If $\hat{\pi}_0 \in (0, 1)$, the optimality condition (12) implies that

$$0 = \hat{\Gamma}_0 = \hat{\mu}_{H,0}^l(r_H^h - r_H^l) - (1 - q)[r_{H,0}^l - (1 - \gamma)\hat{r}_{L,0}^h - \gamma\hat{r}_{L,0}^l]. \quad (\text{A.45})$$

The second component of $\hat{\Gamma}_0$ is pinned down by the given equilibrium path from period 1 onward. This implies that there is a unique value $\hat{\mu}_{H,0}^l$ that solves the equation above. The value of $\hat{\pi}_0$ can then be determined using equation (A.44).

Induction Step. Suppose that the claimed result holds for any $t \leq \tau$ and let $t = \tau + 1$. Consider period τ . Note first that $\hat{\pi}_\tau = 1$ cannot be part of an equilibrium path because $\hat{\mu}_{H,\tau+1}^l = \gamma q$, which cannot hold by assumption.

Suppose next that $\hat{\pi}_\tau \in (0, 1)$. Then the previous argument shows that both $\hat{\mu}_{H,\tau}^l$ and $\hat{\pi}_\tau$ are pinned down by the continuation equilibrium from period $\tau + 1$. The induction argument then applies observing that $\hat{\mu}_{H,\tau}^l = \gamma q$ could arise only for a knife-edge case.

Finally, suppose that $\hat{\pi}_\tau = 0$. Then, $\hat{\mu}_{H,\tau}^l$ is again pinned down by equation (A.44) replacing period 0 with period τ and period 1 with period $\tau + 1$. The equality $\hat{\mu}_{H,\tau}^l = \gamma q$ implies $\hat{\mu}_{H,\tau+1}^l = \gamma(2 - \gamma)q$, which is ruled out by assumption. Thus, if $\hat{\mu}_{H,\tau}^l > \gamma q$, the induction argument applies to period τ . $\square$

The claim that a mixed-strategy steady-state equilibrium is not stable follows from the observation that the value of $\mu_H^{l,ss}$ in a steady state verifies $\mu_H^{l,ss} \in (\gamma q, q)$ and that $\mu_H^{l,ss}$ is equal to $\gamma(2 - \gamma)q$ only in a knife-edge case. For cycles where buyers play a mixed strategy at the peak, note that convergence is ensured if after some time τ , $\mu_{H,\tau}^l = \mu_{H,0}^l$ where $\mu_{H,0}^l \in (\gamma q, \gamma(2 - \gamma)q)$ is the value of the state variable at the peak of the cycle. Claim A.1 implies again that convergence cannot occur.

A.7. Proof of Proposition 3

The existence part of the result follows from the proof of Proposition 2. In particular, the fact that no cycle exists is ensured by the assumption that (C_1) does not hold, which ensures, in turn, that (C_T) does not hold for any T . The stability part of the result follows from the intermediate results in the proof of Corollary 2. Only the liquid steady state and the illiquid steady state are stable because these are the only equilibria in which buyers never randomize.

A.8. Proof of Prediction 2

In the trough of the cycle, the reservation price of an lL asset owner is given by

$$r_{L,t}^{l,(T)} = \begin{cases} u_L^l + \delta r_{L,t+1}^{l,(T)} & \text{if } t = 1, \dots, T - 2 \\ u_L^l + \delta r_H^l & \text{if } t = T - 1. \end{cases} \quad (\text{A.46})$$

The asset price is given by $p_t = r_{L,t}^{l,(T)}$ for $t = 1, \dots, T-1$ and $p_T = r_H^l$. Solving for p_t in the trough of the cycle gives equation (19), from which the following expression for the price change obtains:

$$\Delta p_t^{(T)} = \begin{cases} -\frac{1-\delta^{T-1}}{1-\delta}(u_H^l - u_L^l) & \text{if } t = 1 \\ \delta^{T-t}(u_H^l - u_L^l) & \text{if } t = 2, \dots, T. \end{cases} \quad (\text{A.47})$$

Because the peak price is a given constant and the price change is only negative after the peak, by construction, $\Delta p_1^{(T)} = -\sum_{t=2}^T \Delta p_t^{(T)}$ and so $\mathbb{E}_t[\Delta p_t^{(T)}] = 0$. It also follows from the equation above that $\Delta p_t^{(T)} \geq \Delta p_{t+1}^{(T)} \geq 0$ for all $t = 2, \dots, T$. This last result proves Prediction 4. The price reversal measure is given by

$$\gamma = \text{Cov}[\Delta p_t^{(T)}, \Delta p_{t+1}^{(T)}] = \mathbb{E}[\Delta p_t^{(T)} \Delta p_{t+1}^{(T)}] \quad (\text{A.48a})$$

$$= \frac{1}{T} \sum_{t=1}^T \Delta p_t^{(T)} \Delta p_{t+1}^{(T)} \quad (\text{A.48b})$$

$$< \frac{1}{T} \left(\sum_{t=2}^{T-1} \Delta p_t^{(T)} \Delta p_{t+1}^{(T)} - |\Delta p_1^{(T)}| \Delta p_T^{(T)} \right) \quad (\text{A.48c})$$

$$< \frac{\Delta p_T^{(T)}}{T} \left(\sum_{t=2}^{T-1} \Delta p_t^{(T)} - |\Delta p_1^{(T)}| \right) < 0. \quad (\text{A.48d})$$

The first inequality follows from the fact that the negative term $\Delta p_1^{(T)} \Delta p_2^{(T)}$ is dropped from the sum. The second inequality follows from the result that $\Delta p_T^{(T)} \geq \Delta p_t^{(T)} \geq 0$ for all $t = 2, \dots, T$. The last result follows from the observation that $\Delta p_1^{(T)} = -\sum_{t=2}^T \Delta p_t^{(T)}$ with all the terms in this sum strictly positive.

A.9. Proof of Proposition 4

In this proof, I drop the superscript (2) for the value π_0 of market liquidity at the peak of the cycle. In the first step of the proof, I reexpress condition (23) as a function of π_0 and π^{ss} only. In the second step, I use the equilibrium conditions to derive a relationship between π_0 and π^{ss} and to check whether the condition above holds. Using equation (A.7) to substitute for $\mu_H^{l,ss}$ and equation (A.25) to substitute for $\mu_{H,0}^{l,(2)}$ and $\mu_{H,1}^{l,(2)}$, equation (23) becomes

$$0 \leq \frac{1}{2} \frac{1 + (3 - 2\gamma)(1 - \pi_0)}{1 - (1 - \gamma)^2(1 - \pi_0)} - \frac{1 - \pi^{ss}}{1 - (1 - \gamma)(1 - \pi^{ss})}, \quad (\text{A.49a})$$

$$\Leftrightarrow 0 \leq \frac{(3 - \gamma)(1 - \pi^{ss}) - 1}{1 + (1 - \gamma)(1 + \pi^{ss})} - (1 - \pi_0), \quad (\text{A.49b})$$

$$\Leftrightarrow \pi_0 \leq 2 \frac{2 - \gamma}{1 + (1 - \gamma)(1 + \pi^{ss})} \pi^{ss}. \quad (\text{A.49c})$$

To derive the expression relating π^{ss} and π_0 , consider first the case in which $q \in [q_1, q_2]$, so that $\pi_0 \in (0, 1)$. In this case, buyers randomize at the peak of the cycle. Setting $u_L^h = u_L^l$ in equations (A.16) and (A.29), the indifference condition in a steady state and at the peak of the cycle are, respectively,

$$0 = \Gamma(\pi^{ss}) = \frac{\gamma q}{1 - (1 - \gamma)(1 - \pi^{ss})} (r_H^h - r_H^l) - (1 - q) \frac{u_H^l - u_L}{1 - \delta(1 - \pi^{ss})}, \quad (\text{A.50a})$$

$$0 = \Gamma_0(\pi_0) = \frac{\gamma(2 - \gamma)}{1 - (1 - \gamma)^2(1 - \pi_0)} q(r_H^h - r_H^l) - (1 - q) \frac{1 + \delta}{1 - \delta^2(1 - \pi_0)} (u_H^l - u_L^l). \quad (\text{A.50b})$$

Tedious but simple manipulations yield the result

$$\pi_0 = \frac{(1 + \delta)(2 - \gamma)}{1 - \gamma + \delta + \delta(1 - \gamma)\pi^{ss}} \pi^{ss}, \quad \text{for } q \in [q_1, q_2]. \quad (\text{A.51})$$

Using expression (A.51) to substitute for π_0 in (A.49c), condition (23) for a cycle to dominate is equivalent to

$$0 \leq \frac{(1 + \delta)(2 - \gamma)}{1 - \gamma + \delta + \delta(1 - \gamma)\pi^{ss}} \pi^{ss} - 2 \frac{2 - \gamma}{1 + (1 - \gamma)(1 + \pi^{ss})} \pi^{ss} \quad (\text{A.52a})$$

$$\Leftrightarrow 0 \leq (1 + \delta)[1 + (1 - \gamma)(1 + \pi^{ss})] - 2[1 - \gamma + \delta + \delta(1 - \gamma)\pi^{ss}] \quad (\text{A.52b})$$

$$\Leftrightarrow 0 \leq (1 - \delta)(1 - \gamma)\pi^{ss} + \gamma(1 - \delta). \quad (\text{A.52c})$$

The last inequality holds because $\pi^{ss} \geq 0$. This proves that the cycle dominates the steady state for $q \in [q_1, q_2]$, and thus, that the threshold $\hat{q}$ belongs to $[q_2, \bar{q}_1]$ if it exists.

Consider next the case $q \in [q_2, \bar{q}_1]$. Buyers make a high offer with probability $\pi_0 = 1$ at the peak of the cycle. The analysis above shows that condition (A.49c) is satisfied when $q = q_2$. When $q = \bar{q}_1$, however, condition (A.49c) does not hold because the right-hand side is equal to $\frac{2(2 - \gamma)}{1 + 2(1 - \gamma)} > 1$. The right-hand side of (A.49c) is strictly increasing in π^{ss} , which is itself increasing in q , so there exists a value $\hat{q} \in (q_2, \bar{q}_1)$ such that condition (A.49c) holds if and only if $q \leq \hat{q}$. This concludes the proof.

Appendix B: Equilibrium Stability

To introduce a stability criterion, an equilibrium transition path must first be defined.

DEFINITION B.1 (Equilibrium Path): An equilibrium path with initial state $\mu_{H,0}^l$ is a sequence of asset distribution $\{\mu_{H,t}^l\}_{t \in \mathbb{N} \setminus \{0\}}$, buyer mixing probability $\{\pi_t\}_{t \in \mathbb{N}}$, and reservation prices $\{r_{a,t}^i\}_{a \in \{L,H\}, t \in \mathbb{N}}^{i=L,h}$ such that, for all $t \geq 0$,

- i) π_t satisfies equation (12) given $\{r_{a,t}^i\}_{a=L,H}^{i=L,h}$, and $\mu_{H,t}^l$,
- ii) for $i \in \{L, h\}$, $r_{a,t}^i$ satisfies equations (15a) and (15b), and
- iii) $\mu_{H,t+1}^l$ satisfies the law of motion (4) given $\mu_{H,t}^l$ and π_t .

DEFINITION B.2 (Stability): A stationary equilibrium is locally stable if, for each μ_t^H in the state space $\mathcal{M}$, there exists an open neighborhood O of μ_t^H such that for any $\mu_{H,0}^l$ in O , there is an equilibrium path with initial value $\mu_{H,0}^l$ that converges to this stationary equilibrium.

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Supporting Information

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Appendix S1: Internet Appendix.
Replication Code.